ABB Robotics

Operating manual
Trouble shooting, IRC5

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Operating manual - Trouble shooting, IRC5

Robot Controller

IRC5

M2004

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Revision: K

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Overview of this manual

About this manual

This manual contains information, procedures and descriptions, for trouble shooting IRC5 based robot systems.

Usage

This manual should be used whenever robot operation is interrupted by malfunction, regardless of whether an error event log message is created or not.

Who should read this manual?

This manual is intended for the following personnel:

- Machine and robot operators qualified to perform very basic trouble shooting and reporting to service personnel.
- Programmers qualified to write and change RAPID programs.
- Specialized trouble shooting personnel, usually very experienced service personnel, qualified for methodically isolating, analyzing and correcting malfunctions within the robot system.

Prerequisites

The reader should:

- Have extensive experience in trouble shooting industrial electro-mechanical machinery.
- Have in depth knowledge of the robot system function.
- Be familiar with the actual robot installation at hand, its surrounding equipment and peripherals.

References

Reference:	Document ID:
Product manual - IRC5	3HAC021313-001
Emergency safety information	3HAC027098-001
General safety information	3HAC031045-001
Operating manual - IRC5 with FlexPendant	3HAC16590-1
Operating manual - RobotStudio	3HAC032104-001
Operating manual - Getting started, IRC5 and RobotStudio	3HAC027097-001
Technical reference manual - System parameters	3HAC17076-1
Application manual - MultiMove	3HAC021272-001

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Revisions

Revision	Description
-	First edition.
A	Information has been added. The document has been partly restructured.
B	Information on how to submit error report has been changed. Information on RAPID change logs have been added. Event log messages have been added.
C	Updated Event log messages.
D	Updated Event log messages.
E	Updated Event log messages.
F	Minor corrections. Updated Event log messages.
G	Minor corrections. Updated Event log messages.
H	New information in section Serial Measurement Unit regarding the battery pack. More detailed information about trouble shooting power supplies DSQC 604, 661 and 662. Removed safety I/O signals: DRV1PANCH1, DRV1PANCH2, DRV1SPEED. New drive system introduced. Drive System 04 and Drive System 09 are both described.
J	Released with RobotWare 5.13 The chapter *Safety* updated with: • Updated safety signal graphics for the levels Danger and Warning, see *Safety signals in the manual on page 7*. • New safety labels on the manipulators, see *Safety symbols on the manipulator labels on page 9*. • Updated the graphic in the section *DANGER - Live voltage inside Drive Module! on page 17*. The contents in the following sections were updated: • Corrections regarding drive system information in chapter *Descriptions and background information on page 65* • Restructured the chapters as per the new document startergy. • Updated the graphic in the *Recommended actions* of the section *No voltage in service outlet on page 38*. • Updated the *Possible causes* in the section *Problem starting the FlexPendant on page 40*. • Updated the graphics in the section *LEDs in the Control Module on page 65*. • Updated the graphic in *Possible causes* of the section *Problem releasing Robot brakes on page 50*.
K	Updated Event log messages.

1 Safety

1.1. Safety signals in the manual

Introduction to safety signals

This section specifies all dangers that can arise when doing the work described in this manual. Each danger consists of:

- A caption specifying the danger level (DANGER, WARNING, or CAUTION) and the type of danger.
- A brief description of what will happen if the operator/service personnel **do not** eliminate the danger.
- An instruction on how to eliminate the danger to simplify doing the work.

Danger levels

The table below defines the captions specifying the danger levels used throughout this manual.

Symbol	Designation	Significance
 danger	DANGER	Warns that an accident *will* occur if the instructions are not followed, resulting in a serious or fatal injury and/or severe damage to the product. It applies to warnings that apply to danger with, for example, contact with high voltage electrical units, explosion or fire risk, risk of poisonous gases, risk of crushing, impact, fall from height, etc.
 warning	WARNING	Warns that an accident *may* occur if the instructions are not followed that can lead to serious injury, possibly fatal, and/or great damage to the product. It applies to warnings that apply to danger with, for example, contact with high voltage electrical units, explosion or fire risk, risk of poisonous gases, risk of crushing, impact, fall from height, etc.
 Electrical shock	ELECTRICAL SHOCK	Warns for electrical hazards which could result in severe personal injury or death.
 caution	CAUTION	Warns that an accident may occur if the instructions are not followed that can result in injury and/or damage to the product. It also applies to warnings of risks that include burns, eye injury, skin injury, hearing damage, crushing or slipping, tripping, impact, fall from height, etc. Furthermore, it applies to warnings that include function requirements when fitting and removing equipment where there is a risk of damaging the product or causing a breakdown.

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Symbol	Designation	Significance
 Electrostatic discharge (ESD)	ELECTROSTATIC DISCHARGE (ESD)	Warns for electrostatic hazards which could result in severe damage to the product.
 Note	NOTE	Describes important facts and conditions.
 Tip	TIP	Describes where to find additional information or how to do an operation in an easier way.

1.2. Safety symbols on the manipulator labels

Introduction to labels

This section describes safety symbols used on labels (stickers) on the manipulator.

Symbol are used in combinations on the labels, describing each specific warning. The descriptions in this section are generic, the labels can contain additional information such as values.

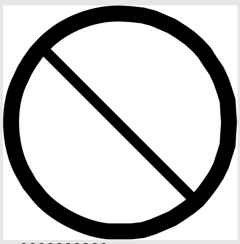
Types of labels

Both the manipulator and the controller are marked with several safety and information labels, containing important information about the product. The information is useful for all personnel handling the manipulator system, for example during installation, service, or operation.

The safety labels are language independent, they only use graphics. See *Symbols on safety labels on page 9*.

The information labels can contain information in text (English, German, and French).

Symbols on safety labels

Symbol	Description
 xx0900000812	**Warning!** Warns that an accident *may* occur if the instructions are not followed that can lead to serious injury, possibly fatal, and/or great damage to the product. It applies to warnings that apply to danger with, for example, contact with high voltage electrical units, explosion or fire risk, risk of poisonous gases, risk of crushing, impact, fall from height, etc.
 xx0900000811	**Caution!** Warns that an accident may occur if the instructions are not followed that can result in injury and/or damage to the product. It also applies to warnings of risks that include burns, eye injury, skin injury, hearing damage, crushing or slipping, tripping, impact, fall from height, etc. Furthermore, it applies to warnings that include function requirements when fitting and removing equipment where there is a risk of damaging the product or causing a breakdown.
 xx0900000839	**Prohibition** Used in combinations with other symbols.

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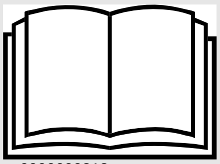
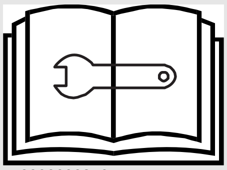
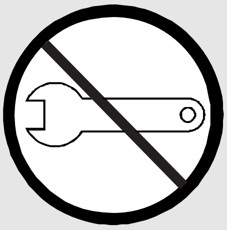
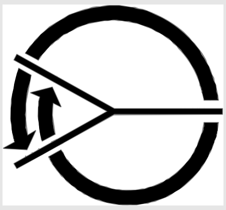
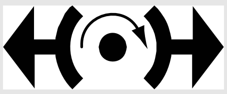
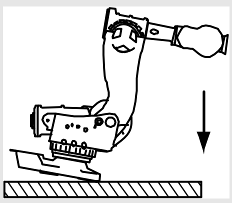
Symbol	Description
 xx0900000813	**Product manual** Read the product manual for details.
 xx0900000816	**Before dismantling see product manual**
 xx0900000815	**Do not dismantle** Dismantling this part can cause injury.
 xx0900000814	**Extended rotation** This axis has extended rotation (working area) compared to standard.
 xx0900000808	**Brake release** Pressing this button will release the brakes. This means that the manipulator arm can fall down.
 xx0900000810	**Tip risk when loosening bolts** The manipulator can tip over if the bolts are not securely fastened.
 xx0900000817	**Crush** Risk for crush injuries.

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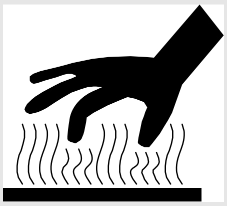
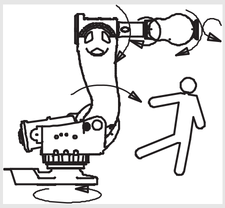
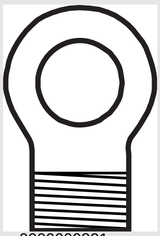
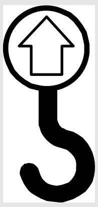
Continued

Symbol	Description
 xx0900000818	**Heat** Risk of heat that can cause burns.
 xx0900000819	**Moving robot** The robot can move unexpectedly.
 xx0900000820	**Brake release buttons**
 xx0900000821	**Lifting bolt**
 xx0900000822	**Lifting of robot**
 xx0900000823	**Oil** Can be used in combination with prohibition if oil is not allowed.

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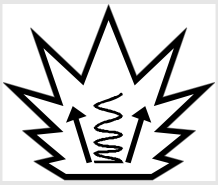
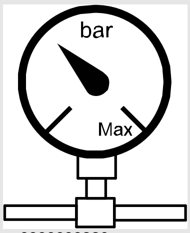
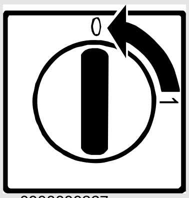
Continued

Symbol	Description
 xx0900000824	**Mechanical stop**
 xx0900000825	**Stored energy** Warns that this part contains stored energy. Used in combination with *Do not dismantle* symbol.
 xx0900000826	**Pressure** Warns that this part is pressurized. Usually contains additional text with the pressure level.
 xx0900000827	**Shut off with handle** Use the power switch on the controller.

1.3. Safety during trouble shooting

General

All normal service work; installation, maintenance and repair work, is usually performed with all electrical, pneumatic and hydraulic power switched off. All manipulator movements are usually prevented by mechanical stops etc.

Trouble shooting work differs from this. While trouble shooting, all or any power may be switched on, the manipulator movement may be controlled manually from the FlexPendant, by a locally running robot program or by a PLC to which the system may be connected.

Dangers during trouble shooting

This implies that special considerations **unconditionally** must be taken when trouble shooting:

- All electrical parts must be considered as *live*.
- The manipulator must at all times be expected to perform any movement.
- Since safety circuits may be disconnected or strapped to enable normally prohibited functions, the system must be expected to perform accordingly.

1.4. Applicable safety standards

Standards, EN ISO

The manipulator system is designed in accordance with the requirements of:

Standard	Description
EN ISO 12100 -1	Safety of machinery - Basic concepts, general principles for design - Part 1: Basic terminology, methodology
EN ISO 12100 -2	Safety of machinery - Basic concepts, general principles for design - Part 2: Technical principles
EN ISO 13849-1	Safety of machinery, safety related parts of control systems - Part 1: General principles for design
EN ISO 13850	Safety of machinery - Emergency stop - Principles for design
EN ISO 10218-1[1]	Robots for industrial environments - Safety requirements -Part 1 Robot
EN ISO 9787	Manipulating industrial robots, Coordinate systems and motion nomenclatures
EN ISO 9283	Manipulating industrial robots, Performance criteria and related test methods
EN ISO 14644-1[2]	Classification of air cleanliness
EN ISO 13732-1	Ergonomics of the thermal environment - Part 1
EN 61000-6-4 (option 129-1)	EMC, Generic emission
EN 61000-6-2	EMC, Generic immunity
EN IEC 60974-1[3]	Arc welding equipment - Part 1: Welding power sources
EN IEC 60974-10[3]	Arc welding equipment - Part 10: EMC requirements
EN 60204-1	Safety of machinery - Electrical equipment of machines - Part 1 General requirements
IEC 60529	Degrees of protection provided by enclosures (IP code)

1. There is a deviation from paragraph 6.2 in that only worst case stop distances and stop times are documented.

2. Only robots with Protection Clean Room.

3. Only valid for arc welding robots. Replaces EN 61000-6-4 for arc welding robots.

European standards

Standard	Description
EN 614-1	Safety of machinery - Ergonomic design principles - Part 1: Terminology and general principles
EN 574	Safety of machinery - Two-hand control devices - Functional aspects - Principles for design
EN 953	Safety of machinery - General requirements for the design and construction of fixed and movable guards

Continues on next page

Other standards

Standard	Description
ANSI/RIA R15.06	Safety Requirements for Industrial Robots and Robot Systems
ANSI/UL 1740 (option 429-1)	Safety Standard for Robots and Robotic Equipment
CAN/CSA Z 434-03 (option 429-1)	Industrial Robots and Robot Systems - General Safety Requirements

1.5 Safe Trouble Shooting

1.5.1. DANGER - Robot without axes' holding brakes are potentially lethal!

Description

Since the robot arm system is quite heavy, especially on larger robot models, it is dangerous if the holding brakes are disconnected, faulty, worn or in any way rendered non-operational.

For instance, a collapsing IRB 7600 arm system may kill or seriously injure a person standing beneath it.

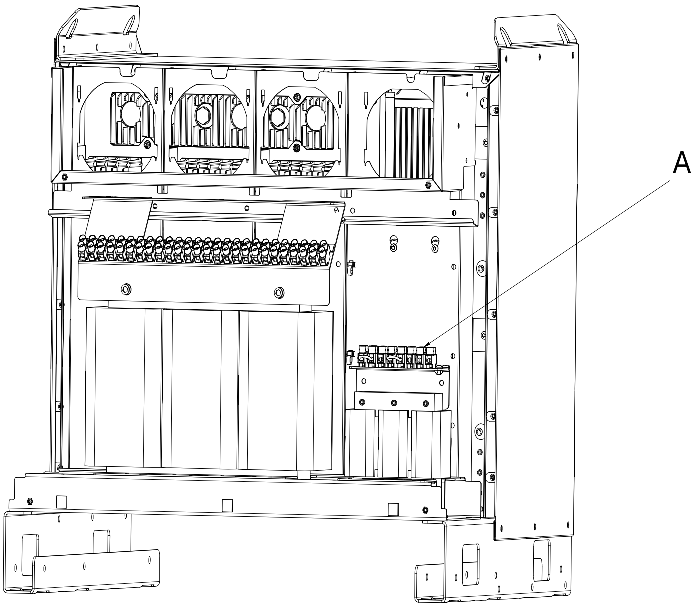
Elimination

	Action	Info/illustration
1.	If you suspect that the holding brakes are non-operational, secure the robot arm system by some other means before working on it.	Weight specifications etc. may be found in the *Product manual* of each robot model.
2.	If you intentionally render the holding brakes non-operational by connecting an external voltage supply, the utmost care must be taken! **DANGER!** **NEVER** stand inside the robot working area when disabling the holding brakes unless the arm system is supported by some other means! **DANGER!** Under no circumstance stand beneath any of the robot's axes!	How to correctly connect an external voltage supply is detailed in the *Product manual* of each robot model.

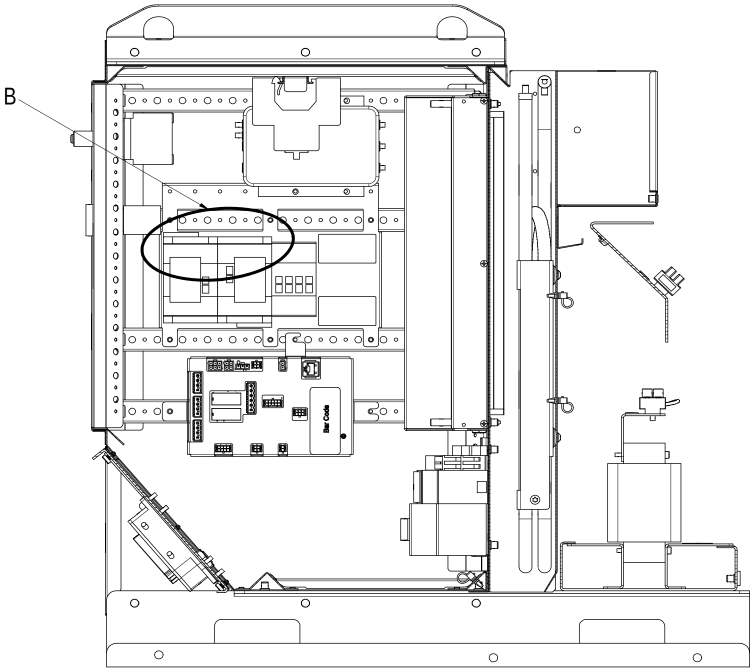
1.5.2. DANGER - Live voltage inside Drive Module!

Description

The Drive Module has live voltage potentially accessible directly behind the rear covers and inside the front cover, even when the main switches have been switched off.



en1000000049



en1000000050

A	Live voltage at transformer terminals **even if the main power switches have been switched off**.
B	Live voltage at Motors ON terminals **even if the main power switches have been switched off**.

Continues on next page

Elimination

Read this information before opening the rear cover of either module.

Step	Action
1.	Make sure the incoming mains power supply has been switched off.
2.	Use a voltmeter to verify that there is not voltage between any of the terminals.
3.	Proceed with the service work.

1.5.3. WARNING - The unit is sensitive to ESD!

Description

ESD (electrostatic discharge) is the transfer of electrical static charge between two bodies at different potentials, either through direct contact or through an induced electrical field. When handling parts or their containers, personnel not grounded may potentially transfer high static charges. This discharge may destroy sensitive electronics.

Elimination

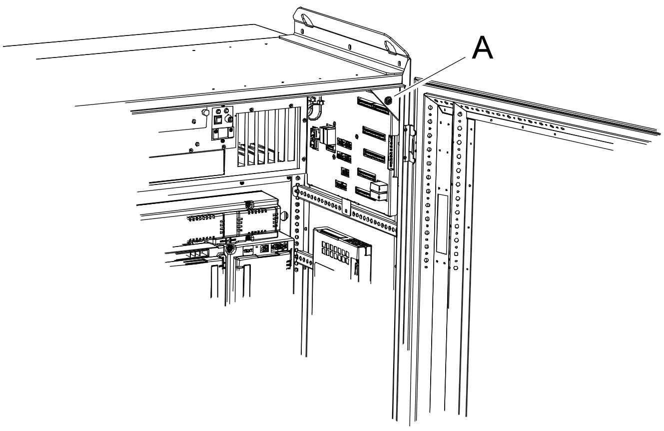
	Action	Note
1.	Use a wrist strap	Wrist straps must be tested frequently to ensure that they are not damaged and are operating correctly.
2.	Use an ESD protective floor mat.	The mat must be grounded through a current-limiting resistor.
3.	Use a dissipative table mat.	The mat should provide a controlled discharge of static voltages and must be grounded.

Location of wrist strap button

The location of the wrist strap button is shown in the following illustration.

IRC5

The wrist strap button is located in the top right corner.



xx0500002171

A	Wrist strap button

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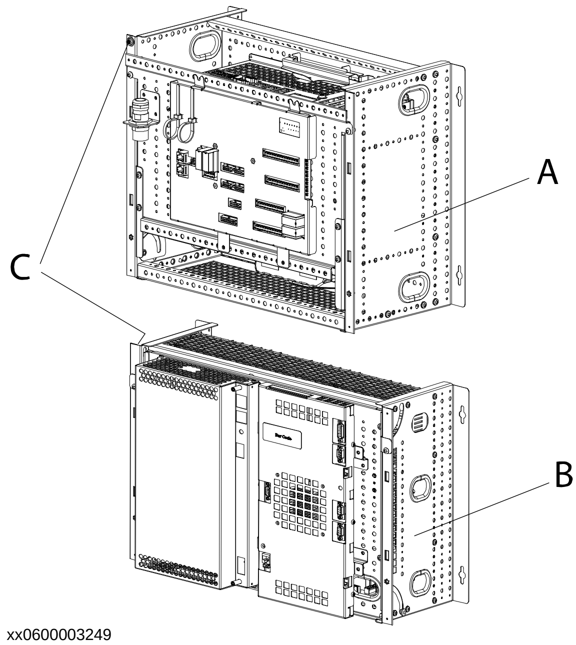
Panel Mounted Controller



xx0600003249

A	Panel Mounted Control Module
B	Panel Mounted Drive Module
C	Wrist strap button NOTE! When not used, the wrist strap must always be attached to the wrist strap button.

1.5.4. CAUTION - Hot parts may cause burns!

Description

During normal operation, many manipulator parts become hot, especially the drive motors and gears. Sometimes areas around these parts also become hot. Touching these may cause burns of various severity.

Because of a higher environment temperature, more surfaces on the manipulator get hot and may result in burns.

NOTE!

The drive parts in the cabinet can be hot.

Elimination

The instructions below detail how to avoid the dangers specified above:

	Action	Info
1.	Always use your hand, at some distance, to feel if heat is radiating from the potentially hot component before actually touching it.	
2.	Wait until the potentially hot component has cooled if it is to be removed or handled in any other way.	
3.	The Bleeder can be hot upto 80 degrees.	

1.5.4. CAUTION - Hot parts may cause burns!

2 Trouble shooting Overview

2.1. Documentation and references

General

A great deal of effort was put into writing the event log messages as well as the technical documentation. Though imperfect, they may give vital clues. They are also constantly being upgraded.

The product documentation is available in several languages.

Read the documentation!

Do not wait until nothing else works to read the manual!

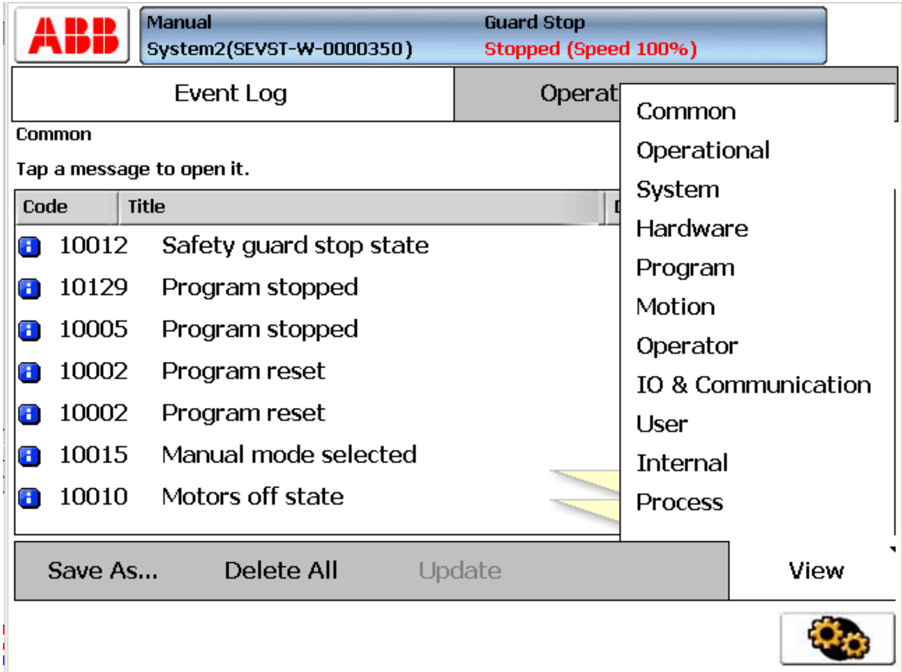
References to document numbers are specified in the chapter *Reference information* in *Product manual - IRC5*.

Read the circuit diagram!

The complete electrical circuitry of the controller is documented in *Product manual - IRC5*. It contains a lot of information useful, or even essential, to a trained trouble shooter.

Read the logs!

The error event logs which may be viewed on either the FlexPendant or RobotStudio, contain lots of information about any malfunction detected by the system.



en0300000547

Check the electronical unit's LEDs!

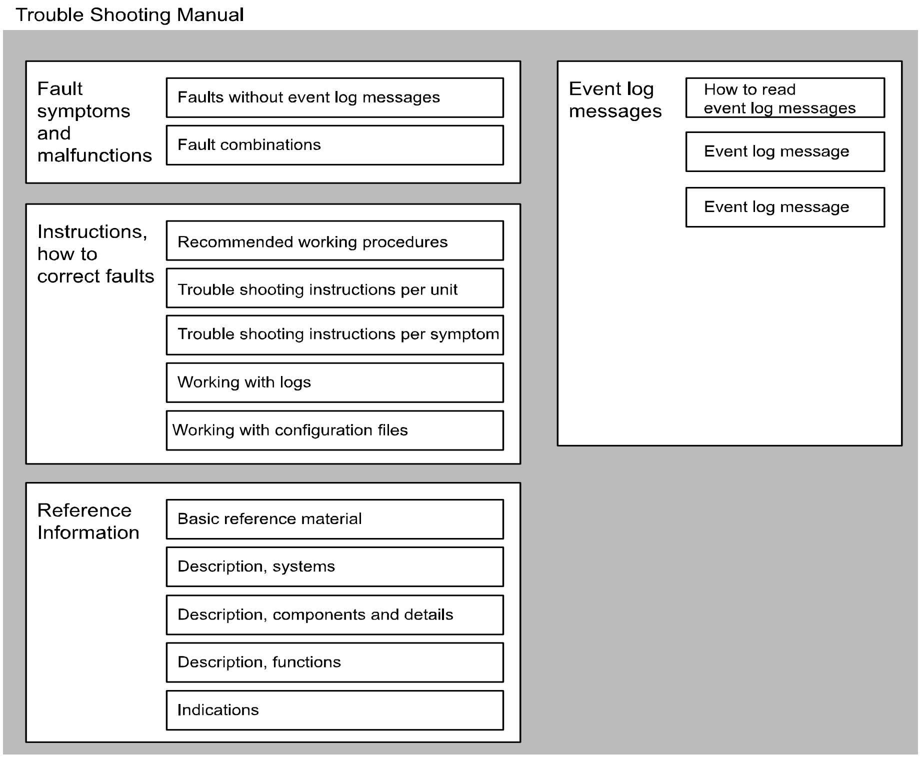
If a fault is thought to be caused by an electronic unit (circuit board in the controller or other), the LEDs on the unit front may give leads.

These are described in section *Indications on page 65*.

2.2. Overview

How to use this manual when trouble shooting

The illustration and description detail how to put the information in this manual to best use during trouble shooting the robot system.



en0400001200

Trouble shooting manual

Fault symptoms and malfunctions:

- Each fault or error is first detected as a symptom, for which an error event log message may or may not be created. It could be an error event log message on the FlexPendant, an observation that the gearbox on axis 6 is getting hot or that the controller can not be started. The faults displaying an event log message are listed in the end of this manual.

Instructions, how to correct faults:

- The instructions are divided into two main categories: descriptions of how to correctly handle the different parts of the system and instructions of how to remedy faults causing the symptoms specified above. The latter category is divided into two subcategories, depending on whether to trouble shoot a specific symptom or a suspected unit causing the problem. The first category contains information on how to use the event log to facilitate trouble shooting, etc.

Recommended working procedures:

- Here, you will find a procedure for how to correctly perform certain specific tasks. These may be used to make sure the seemingly irrational behavior of the system is not due to incorrect handling.

Basic reference info:

- This section contains information about what tools to use, references to documents that may be useful when trouble shooting, etc.

Description, systems:

- The different systems and sub-systems are described to give a better understanding of its function when it works "as it's supposed to". This enables the trouble shooter to better see and understand the differences between a system that's functional and one that's not.

Description, components and details:

- Specific details of the system are described with regards to their function, etc.

Description, functions:

- Contains descriptions on how specific functions within the system work, e.g the RUN chain, and what signals and other systems affect that particular function. This provides for a better understanding of the relations and mechanisms of the robot system.

Indications

- All indication LEDs and other indications (as found on the Control and Drive Modules as well as separate circuit boards, etc) are described in this section along with information about their indication modes and significances respectively. Recommended actions are often specified or references containing such instructions.

Event log messages:

- This section is basically a printout of all available event log messages. These may be displayed either on the FlexPendant or using RobotStudio. Having access to all messages at the same time may be useful during trouble shooting.

Additional information

In addition to the information given in this document, other documents may provide vital information, e.g. the Circuit Diagram.

Such useful documents are listed in *Overview of this manual on page 5*.

Continues on next page

2.3. Standard toolkit

General

Listed are tools required to perform the actual trouble shooting work. All tools required to perform any corrective measure, such as replacing parts, are listed in their Product Manual section respectively.

Contents, standard toolkit, IRC5

Tool	Remark
Screw driver, Torx	Tx10
Screw driver, Torx	Tx25
Ball tipped screw driver, Torx	Tx25
Screw driver, flat blade	4 mm
Screw driver, flat blade	8 mm
Screw driver, flat blade	12 mm
Screw driver	Phillips-1
Box spanner	8 mm

Contents, standard toolkit, trouble shooting

Qty	Art. no.	Tool	Rem.
-	-	Normal shop tools	Contents as specified above.
1	-	Multimeter	-
1	-	Oscilloscope	-
1	-	Recorder	-

2.4 Tips and Tricks while trouble shooting

2.4.1. Trouble shooting strategies

Isolate the fault!

Any fault may give rise to a number of symptoms, for which error event log messages may or may not be created. In order to effectively eliminate the fault, it is vital to distinguish the original symptom from the consequential ones.

A help in isolating the fault may be creating a historical fault log as specified in section *Make a historical fault log! on page 29*.

Split the fault chain in two!

When trouble shooting any system, a good practice is to split the fault chain in two. This means:

- identify the complete chain.
- decide and measure the expected value at the middle of the chain.
- use this to determine in which half the fault is caused.
- split this half into two new halves, etc.
- finally, a single component may be isolated. The faulty one.

Example

A specific IRB 7600 installation has a 12 VDC power supply to a tool at the manipulator wrist. This tool does not work, and when checked, there is no 12 VDC supply to it.

- Check at the manipulator base to see if there is 12 VDC supply. Measurement show there are *no* 12 VDC supply. (Reference: Circuit Diagram in the *Product manual, IRC5*)
- Check any connector between the manipulator and the power supply in the controller. Measurement show there are *no* 12 VDC supply. (Reference: Circuit Diagram in the *Product manual, IRC5*)
- Check the power supply unit LED. (Reference: *Indications on page 65*)

Check communication parameters and cables!

The most common causes of errors in serial communication are:

- Faulty cables (e.g. send and receive signals are mixed up)
- Transfer rates (baud rates)
- Data widths that are incorrectly set.

Check the software versions!

Make sure the RobotWare and other software run by the system are the correct version. Certain versions are not compatible with certain hardware combinations.

Also, make a note of all software versions run, since this will be useful information to the ABB support people.

How to file a complete error report to your local ABB service personnel is detailed in section *Filing an error report on page 30*.

2.4.2. Work systematically

Do not replace units randomly!

Before replacing *any part at all*, it is important to establish a probable cause for the fault, thus determining which unit to replace.

Randomly replacing units may sometimes solve the acute problem, but also leaves the trouble shooter with a number of units that may/may not be perfectly functional.

Replace one thing at a time!

When replacing a presumably faulty unit that has been isolated, it is important that **only one** unit be replaced at a time.

Always replace components as detailed in the Repairs section of the Product manual of the robot or controller at hand.

Test the system after replacing to see if the problem has been solved.

If replacing several units at once:

- it is impossible to determine which of the units was causing the fault.
- it greatly complicates ordering a new spare part.
- it may introduce new faults to the system.

Take a look around!

Often, the cause may be evident once you see it. In the area of the unit acting erroneously, be sure to check:

- Are the attachment screws secured?
- Are all connectors secured?
- Are all cabling free from damage?
- Are the units clean (especially for electronic units)?
- Is the correct unit fitted?

Check for tools left behind!

Some repair and maintenance work require using special tools to be fitted to the robot equipment. If these are left behind (e.g. balancing cylinder locking device or signal cable to a computer unit used for measuring purposes), they may cause erratic robot behavior.

Make sure all such tools are removed when maintenance work is complete!

2.4.3. Keeping track of history

Make a historical fault log!

In some cases, a particular installation may give rise to faults not encountered in others. Therefore, charting each installation may give tremendous assistance to the trouble shooter.

To facilitate trouble shooting, a log of the circumstances surrounding the fault gives the following advantages:

- it enables the trouble shooter to see patterns in causes and consequences not apparent at each individual fault occurrance.
- it may point out a specific event always taking place just before the fault, for example a certain part of the work cycle being run.

Check up the history!

Make sure you always consult the historical log if it is used. Also remember to consult the operator, or similar, who was working when the problem first occurred.

At what stage did the fault occur?

What to look for during trouble shooting depends greatly of when the fault occurred: was the robot just freshly installed? Was it recently repaired?

The table gives specific hints to what to look for in specific situations:

If the system has just:	then:
been installed	Check: • the configuration files • connections • options and their configuration
been repaired	Check: • all connections to the replaced part • power supplies • that the correct part has been fitted
had a software upgrade	Check: • software versions • compatibilities between hardware and software • options and their configuration
been moved from one site to another (an already working robot)	Check: • connections • software versions

2.5. Filing an error report

Introduction

If you require the assistance of ABB support personnel in trouble shooting your system, you may file a formal error report as detailed below.

In order for the ABB support personnel to better solve your problem, you may attach a special diagnostics file that the system generates on demand.

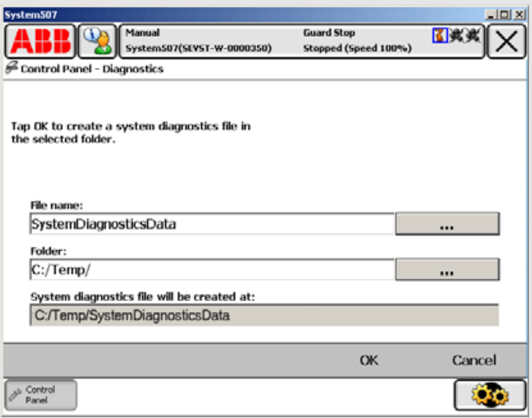
The diagnostics file includes:

- **Event log** A list of all system events.
- **Backup** A backup of the system taken for diagnostics purposes.
- **System information** Internal system information useful to ABB support personnel.

NOTE that it is not required to create or attach any additional files to the error report if not explicitely requested by the support personnel!

Creating the diagnostics file

The diagnostics file is created manually as detailed below.

	Action
1.	Tap **ABB**, then **Control Panel** and then **Diagnostics**. A display is shown:  en0500002175
2.	Specify the name you want for the diagnostics file, the save folder of it and tap **OK**. The default save folder is C:/Temp, but any folder may be selected, for instance an externally connected USB memory. This may take a couple of minutes, while “Creating file. Please wait!” is displayed.
3.	To shorten file transfer time, you may compress the data into a zip-file.
4.	Write a regular e-mail addressed to your local ABB support personnel, and make sure to include the following information: • Robot serial number • RobotWare version • External options • A written fault description. The more detailed, the easier for the ABB support personnel to assist you. • if available, enclose the license key. • attach the diagnostics file!
5.	Mail it!

3 Troubleshooting by fault symptoms

3.1. Start-up failures

Introduction

This section describes possible faults during start-up and the recommended action for each failure.

Consequences

Problem starting the sytem.

Symptoms and causes

The following are the possible symptoms of a start-up failure:

- LEDs not lit on any unit.
- Earth fault protection trips.
- Unable to load the system software.
- FlexPendant not responding.
- FlexPendant starts, but does not respond to any input.
- Disk containing the system software does not start correctly.

Recommended actions

The following are the recommended actions to be taken during a start-up failure:

NOTE!

This may be due to a loss of power supply in many stages.

	Action	Info/illustration
1.	Make sure the main power supply to the system is present and is within the specified limits.	Your plant or cell documentation can provide this information.
2.	Make sure that the main transformer in the Drive module is correctly connected to the mains voltage levels at hand.	How to strap the mains transformer is detailed in the product manual for the controller.
3.	Make sure that the main switches are switched on.	
4.	Make sure that the power supply to the Control module and Drive module are within the specified limits.	If required, trouble shoot the power supply units as explained in section *Trouble shooting power supply on page 56*.
5.	If no LEDs lit, proceed to section *All LEDs are OFF at Controller on page 36*.	
6.	If the system is not responding, proceed to section *Controller not responding on page 33*.	
7.	If the FlexPendant is not responding, proceed to section *Problem starting the FlexPendant on page 40*.	

Continues on next page

Continued

	Action	Info/illustration
8.	If the FlexPendant starts, but does not communicate with the controller, proceed to section *Problem connecting FlexPendant to the controller on page 41*.	

3.2. Controller not responding

Description

This section describes the possible faults and the recommended actions for each failure:

- Robot controller not responding
- LED indicators not lit

Consequences

System cannot be operated using the FlexPendant.

Possible causes

	Symptoms	Recommended action
1	Controller not connected to the mains power supply.	Ensure that the mains power supply is working and the voltage level matches that of the controller requirement.
2	Main transformer is malfunctioning or not connected correctly.	Ensure that the main transformer is connected correctly to the mains voltage level.
3	Main fuse (Q1) might have tripped.	Ensure that the mains fuse (Q1) inside the Drive Module is not tripped
4	Connection missing between the Control and Drive modules.	If the Drive Module does not start although the Control Module is working and the Drive Module main switch has been switched on, ensure that all the connections between the Drive module and the Control module are connected correctly.

3.3. Low Controller performance

Description

The controller performance is low, and seems to work irrationally.

The controller is *not* completely“ dead”. If it is, proceed as detailed in section *Controller not responding on page 33*.

Consequences

These symptoms can be observed:

- Program execution is sluggish, seemingly irrational and sometimes stalls.

Possible causes

The computer system is experiencing too high load, which may be due to one, or a combination, of the following:

- Programs containing too high a degree of logical instructions only, causing too fast program loops and in turn, overloads the processor.
- The I/O update interval is set to a low value, causing frequent updates and a high I/O load.
- Internal system cross connections and logical functions are used too frequently.
- An external PLC, or other supervisory computer, is addressing the system too frequently, overloading the system.

Recommended actions

	Action	Info/illustration
1.	Check whether the program contains logical instructions (or other instructions that take “no time” to execute), since such programs may cause the execution to loop if no conditions are fulfilled. To avoid such loops, you can test by adding one or more WAIT instructions. Use only short WAIT times, to avoid slowing the program down unnecessarily.	Suitable places to add WAIT instructions can be: • In the main routine, preferably close to the end. • In a WHILE/FOR/GOTO loop, preferably at the end, close to the ENDWHILE/ENDFOR etc. part of the instruction.
2.	Make sure the I/O update interval value for each I/O board is not too low. These values are changed using RobotStudio. I/O units that are not read regularly may be switched to “change of state” operation as detailed in the RobotStudio manual.	ABB recommends these poll rates: • DSQC 327A: 1000 • DSQC 328A: 1000 • DSQC 332A: 1000 • DSQC 377A: 20-40 • All others: >100
3.	Check whether there is a large amount of cross connections or I/O communication between PLC and robot system.	Heavy communication with PLCs or other external computers can cause heavy load in the robot system main computer.

	Action	Info/illustration
4.	Try to program the PLC in such a way that it uses event driven instructions, instead of looped instructions.	The robot system have a number of fixed system inputs and outputs that may be used for this purpose. Heavy communication with PLCs or other external computers can cause heavy load in the robot system main computer.

3.4. All LEDs are OFF at Controller

Description

No LEDs at all are lit on the Control Module or the Drive Module respectively.

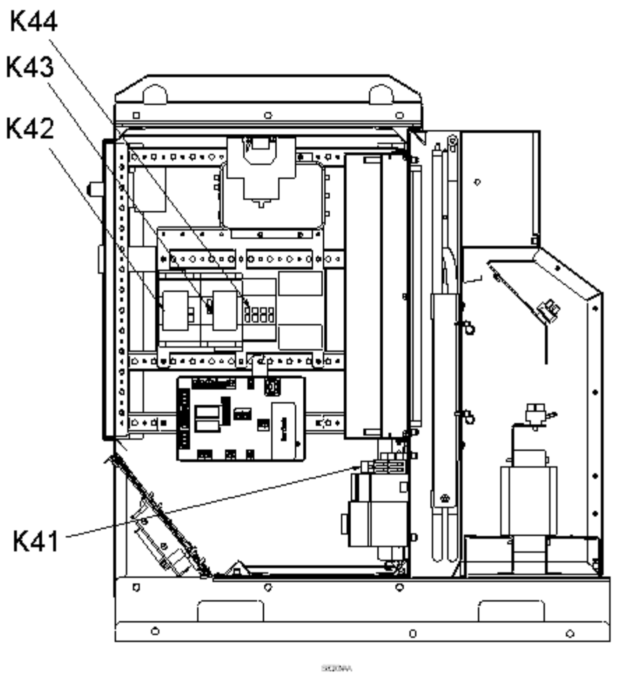
Consequences

The system cannot be operated or started at all.

Possible causes

The symptom can be caused by (the causes are listed in order of probability):

- The system is not supplied with power.
- The main transformer is not connected for the correct mains voltage.
- Circuit breaker F6 (if used) is malfunctioning or open for any other reason.
- Contactor K41 is malfunctioning or open for any other reason.



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Recommended actions

	Action	Info
1.	Make sure the main switch has been switched on.	
2.	Make sure the system is supplied with power.	Use a voltmeter to measure incoming mains voltage.
3.	Check the main transformer connection.	The voltages are marked on the terminals. Make sure they match the shop supply voltage.
4.	Make sure circuit breaker F6 (if used) is closed in position 3.	The circuit breaker F6 is shown in the circuit diagram in the product manual for the controller.
5.	Make sure contactor K41 opens and closes when ordered.	
6.	 - Disconnect connector X1 from the Drive Module power supply and measure the incoming voltage.	Measure between pins X1.1 and X1.5.
7.	If the power supply incoming voltage is correct (230 VAC) but the LEDs still do not work, replace the Drive Module power supply.	Replace the power supply as detailed in the product manual for the controller.

3.5. No voltage in service outlet

Description

Some Control Modules are equipped with service voltage outlet sockets, and this information applies to these modules only.

No voltage is available in the Control Module service outlet for powering external service equipment.

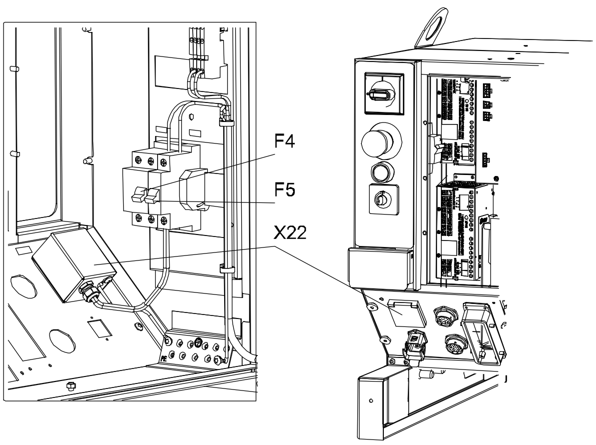
Consequences

Equipment connected to the Control Module service outlet does not work.

Probable causes

The symptom can be caused by (the causes are listed in order of probability):

- Tripped circuit breaker (F5)
- Tripped earth fault protection (F4)
- Mains power supply loss
- Transformers incorrectly connected



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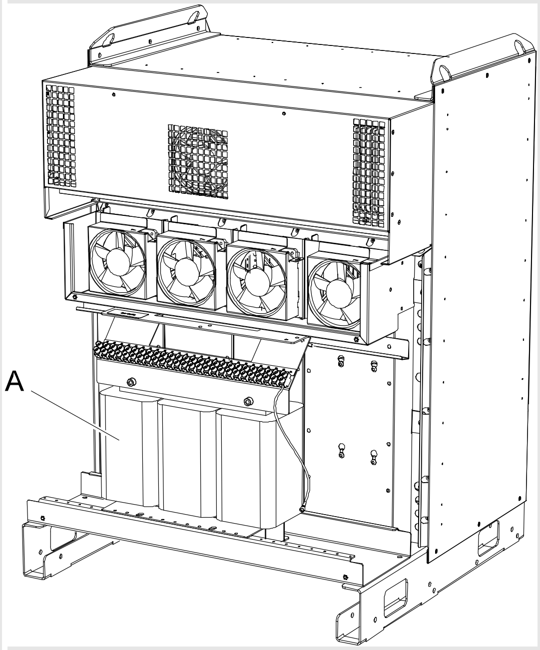
Recommended actions

	Action	Info
1.	Make sure the circuit breaker in the Control Module has not been tripped.	Make sure any equipment connected to the service outlet does not consume too much power, causing the circuit breaker to trip.
2.	Make sure the earth fault protection has not been tripped.	Make sure any equipment connected to the service outlet does not conduct current to ground, causing the earth fault protection to trip.
3.	Make sure the power supply to the robot system is within specifications.	Refer to the plant documentation for voltage values.
4.	Make sure the transformer (A) supplying the outlet is correctly connected, i.e. input and output voltages in accordance with specifications.	 xx0500002028 Refer to the plant documentation for voltage values.

3.6. Problem starting the FlexPendant

Description

The FlexPendant is completely or intermittently "dead".

No entries are possible, and no functions are available.

If the FlexPendant starts but does not display a screen image, proceed as detailed in section *Problem connecting FlexPendant to the controller on page 41*.

Consequences

The system cannot be operated using the FlexPendant.

Possible causes

The symptom can be caused by (the causes are listed in order of probability):

- The system has not been switched on.
- The FlexPendant is not connected to the controller.
- The cable from the controller is damaged.
- The cable connector is damaged.
- FlexPendant power supply from controller is faulty.

Recommended actions

The following actions are recommended (listed in order of probability):

	Action	Info
1.	Make sure the system is switched on and that the FlexPendant is connected to the controller.	How to connect the FlexPendant to the controller is detailed in *Operating manual - Getting started, IRC5 and RobotStudio*.
2.	Inspect the FlexPendant cable for any visible damage.	If faulty, replace the FlexPendant.
3.	If possible, test by connecting a different FlexPendant to eliminate the FlexPendant and cable as error sources.	
4.	If possible, test the FlexPendant with a different controller to eliminate the controller as error source.	

Continues on next page

3.7. Problem connecting FlexPendant to the controller

Description

The FlexPendant starts but does not display a screen image.

No entries are possible, and no functions are available.

The FlexPendant is not completely dead. If it is dead, proceed as detailed in section *Problem starting the FlexPendant on page 40*.

Consequences

The system cannot be operated using the FlexPendant.

Possible causes

The symptom can be caused by (the causes are listed in order of probability):

- The Ethernet network has problems.
- The main computer has problems.

Recommended actions

The following actions are recommended (listed in order of probability):

	Action	Info
1.	Check all cables from power supply unit to main computer, making sure these are correctly connected.	
2.	Make sure the FlexPendant has been correctly connected to the controller.	
3.	Check all indication LEDs on all units in the controller.	All indication LEDs and their significance are specified in section *Indications on page 65*.
4.	Check all status signals on the main computer.	

3.8. Erratic event messages on FlexPendant

Description

The event messages displayed on the FlexPendant are erratic and do not seem to correspond to any actual malfunctions on the robot. Several types of messages can be displayed, seemingly erroneously.

This type of fault may occur after major manipulator disassembly or overhaul, if not performed correctly.

Consequences

Major operational disturbances due to the constantly appearing messages.

Possible causes

The symptom can be caused by (the causes are listed in order of probability):

- Internal manipulator cabling not correctly performed. Causes may be: faulty connection of connectors, cable loops too tight causing the cabling to get strained during manipulator movements, cable insulation chafed or damaged by rubbing short-circuiting signals to earth.

Recommended actions

The following actions are recommended (listed in order of probability):

	Action	Info
1.	Inspect all internal manipulator cabling, especially all cabling disconnected, connected re-routed or bundled during recent repair work.	Refit any cabling as detailed in the product manual for the robot.
2.	Inspect all cable connectors to make sure these are correctly connected and tightened.	
3.	Inspect all cable insulation for damage.	Replace any faulty cabling as detailed in the product manual for the robot.

3.9. Problem jogging the robot

Description

The system can be started but the joystick on the FlexPendant does not work.

Consequences

The robot can not be jogged manually.

Possible causes

The symptom can be caused by (the causes are listed in order of probability):

- The joystick is malfunctioning.
- The joystick may be deflected.

Recommended actions

The following actions are recommended (listed in order of probability):

	Action	Info
1.	Make sure the controller is in manual mode.	How to change operating mode is described in *Operating manual - IRC5 with FlexPendant*.
2.	Make sure the FlexPendant is connected correctly to the Control Module.	
3.	Reset the FlexPendant.	Press Reset button located on the back of the FlexPendant. **NOTE!** The Reset button resets the FlexPendant not the system on the Controller.

3.10. Reflashing firmware failure

Description

When reflashing firmware, the automatic process can fail.

Consequences

The automatic reflashing process is interrupted and the system stops.

Possible causes

This fault usually occurs due to a lack of compatibility between hardware and software.

Consequences

The following actions are recommended (listed in order of probability):

	Action	Info
1.	Check the event log for a message specifying which unit failed.	The logs may also be accessed from RobotStudio.
2.	Was the relevant unit recently replaced? If YES; make sure the versions of the old and new unit is identical. If NO; check the software versions.	
3.	Was the RobotWare recently replaced? If YES; make sure the versions of the old and new unit is identical. If NO; proceed below!	
4.	Check with your local ABB representative for a firmware version compatible with your hardware/software combination.	

3.11. Inconsistent path accuracy

Description

The path of the robot TCP is not consistent. It varies from time to time, and is sometimes accompanied by noise emerging from bearings, gearboxes, or other locations.

Consequences

Production is not possible.

Possible causes

The symptom can be caused by (the causes are listed in order of probability):

- Robot not calibrated correctly.
- Robot TCP not correctly defined.
- Parallel bar damaged (applies to robots fitted with parallel bars only).
- Mechanical joint between motor and gearbox damaged. This often causes noise to be emitted from the faulty motor.
- Bearings damaged or worn (especially if the path inconsistency is coupled with clicking or grinding noises from one or more bearings).
- The wrong robot type may be connected to the controller.
- The brakes may not be releasing correctly.

Recommended actions

In order to remedy the symptom, the following actions are recommended (the actions are listed in order of probability):

	Action	Info/Illustration
1.	Make sure the robot tool and work object are correctly defined.	How to define these are detailed in *Operating manual - IRC5 with FlexPendant*.
2.	Check the revolution counters' positions.	Update if required.
3.	If required, recalibrate the robot axes.	How to calibrate the robot is detailed in *Operating manual - IRC5 with FlexPendant*.
4.	Locate the faulty bearing by tracking the noise.	Replace faulty bearing as specified in the product manual for the robot.
5.	Locate the faulty motor by tracking the noise. Study the path of the robot TCP to establish which axis, and thus which motor, may be faulty.	Replace the faulty motor/gearbox as specified in the product manual for the robot.
6.	Check the trueness of the parallel bar (applies to robots fitted with parallel bars only).	Replace the faulty parallel bar as specified in the product manual for the robot.
7.	Make sure the correct robot type is connected as specified in the configuration files.	
8.	Make sure the robot brakes work properly.	Proceed as detailed in section *Problem releasing Robot brakes on page 50*.

3.12. Oil and grease stains on motors and gearboxes

Description

The area surrounding the motor or gearbox shows signs of oil leaks. This can be at the base, closest to the mating surface, or at the furthest end of the motor at the resolver.

Consequences

Besides the dirty appearance, in some cases there are no serious consequences if the leaked amount of oil is very small. **However**, in some cases the leaking oil lubricates the motor brake, causing the manipulator to collapse at power down.

Possible causes

The symptom can be caused by (the causes are listed in order of probability):

- Leaking seal between gearbox and motor.
- Gearbox overfilled with oil.
- Gearbox oil too hot.

Recommended actions

In order to remedy the symptom, the following actions are recommended (the actions are listed in order of probability):

	Action	Info
1.	**CAUTION!** Before approaching the potentially hot robot component, observe the safety information in section *CAUTION - Hot parts may cause burns! on page 21*.	
2.	Inspect all seals and gaskets between motor and gearbox. The different manipulator models use different types of seals.	Replace seals and gaskets as specified in the product manual for the robot.
3.	Check the gearbox oil level.	Correct oil level is specified in the product manual for the robot.
4.	Too hot gearbox oil may be caused by: • Oil quality or level used is incorrect. • The robot work cycle runs a specific axis too hard. Investigate whether it is possible to program small "cooling periods" into the application. • Overpressure created inside gearbox.	Check the recommended oil level and type as specified in the product manual for the robot. Manipulators performing certain, extremely heavy duty work cycles may be fitted with vented oil plugs. These are not fitted to normal duty manipulators, but may be purchased from your local ABB representative.

3.13. Mechanical noise

Description

During operation, no mechanical noise should be emitted from motors, gearboxes, bearings, or similar. A faulty bearing often emits scraping, grinding, or clicking noises shortly before failing.

Consequences

Failing bearings cause the path accuracy to become inconsistent, and in severe cases, the joint can seize completely.

Possible causes

The symptom can be caused by (the causes are listed in order of probability):

- Worn bearings.
- Contaminations have entered the bearing races.
- Loss of lubrication in bearings.

If the noise is emitted from a gearbox, the following can also apply:

- Overheating.

Recommended actions

The following actions are recommended (listed in order of probability):

	Action	Info
1.	**CAUTION!** Before approaching the potentially hot robot component, observe the safety information in section *CAUTION - Hot parts may cause burns! on page 21*.	
2.	Determine which bearing is emitting the noise.	
3.	Make sure the bearing has sufficient lubrication.	As specified in the product manual for the robot.
4.	If possible, disassemble the joint and measure the clearance.	As specified in the product manual for the robot.
5.	Bearings inside motors are not to be replaced individually, but the complete motor is replaced.	Replace faulty motors as specified in the product manual for the robot.
6.	Make sure the bearings are fitted correctly.	Also see the product manual for the robot for general instruction on how to handle bearings.

	Action	Info
7.	Too hot gearbox oil may be caused by: • Oil quality or level used is incorrect. • The robot work cycle runs a specific axis too hard. Investigate whether it is possible to program small "cooling periods" into the application. • Overpressure created inside gearbox.	Check the recommended oil level and type as specified in the product manual for the robot. Manipulators performing certain, extremely heavy duty work cycles may be fitted with vented oil plugs. These are not fitted to normal duty manipulators, but may be purchased from your local ABB representative.

3.14. Manipulator crashes on power down

Description

The manipulator is able to work correctly while Motors ON is active, but when Motors OFF is active, it collapses under its own weight.

The holding brake, integral to each motor, is not able to hold the weight of the manipulator arm.

Consequences

The fault can cause severe injuries or death to personnel working in the area or severe damage to the manipulator and/or surrounding equipment.

Possible causes

The symptom can be caused by (the causes are listed in order of probability):

- Faulty brake.
- Faulty power supply to the brake.

Recommended actions

The following actions are recommended (listed in order of probability):

	Action	Info
1.	Determine which motor(s) causes the robot to collapse.	
2.	Check the brake power supply to the collapsing motor during the Motors OFF state.	Also see the circuit diagrams in the product manuals for the robot and the controller .
3.	Remove the resolver of the motor to see if there are any signs of oil leaks.	If found faulty, the motor must be replaced as a complete unit as detailed in the product manual for the robot.
4.	Remove the motor from the gearbox to inspect it from the drive side.	If found faulty, the motor must be replaced as a complete unit as detailed in the product manual for the robot.

Continues on next page

3.15. Problem releasing Robot brakes

Description

When starting robot operation or jogging the robot, the internal robot brakes must release in order to allow movements.

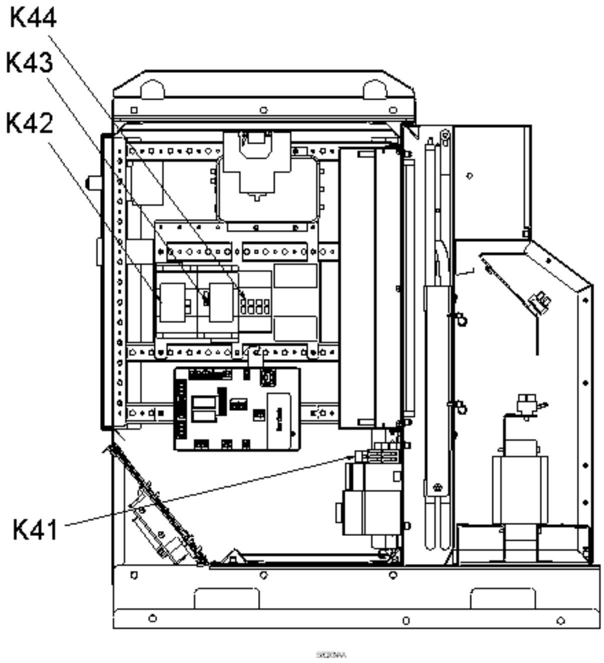
Consequences

If the brakes do not release, no robot movement is possible, and a number of error log messages can occur.

Possible causes

The symptom can be caused by (the causes are listed in order of probability):

- Brake contactor (K44) does not work correctly.
- The system does not go to status Motors ON correctly.
- Faulty brake on the robot axis.
- Supply voltage 24V BRAKE missing.



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Recommended actions

This section details how to proceed when the robot brakes do not release.

	Action	Info
1.	Make sure the brake contactor is activated.	A ‘tick’ should be audible, or you may measure the resistance across the auxiliary contacts on top of the contactor.
2.	Make sure the RUN contactors (K42 and K43) are activated. NOTE that **both** contactors must be activated, not just one!	A ‘tick’ should be audible, or you may measure the resistance across the auxiliary contacts on top of the contactor.
3.	Use the push buttons on the robot to test the brakes. If just one of the brakes malfunctions, the brake at hand is probably faulty and must be replaced. If none of the brakes work, there is probably no 24V BRAKE power available.	The location of the push buttons differ, depending on robot model. Please refer to the product manual for the robot!
4.	Check the Drive Module power supply to make sure 24V BRAKE voltage is OK.	
5.	A number of other faults within the system can cause the brakes to remain activated. In such cases, event log messages will provide additional information.	The event log messages can also be accessed using RobotStudio.

3.16. Intermittent errors

Description

During operation, errors and malfunctions may occur, in a seemingly random way.

Consequences

Operation is interrupted, and occasionally, event log messages are displayed, that sometimes do not seem to be related to any actual system malfunction. This sort of problem sometimes affects the Emergency stop or Enable chains respectively, and may at times be very hard to pinpoint.

Probable causes

Such errors may occur anywhere in the robot system and may be due to:

- external interference
- internal interference
- loose connections or dry joints, e.g. incorrectly connected cable screen connections.
- thermal phenomena , e.g. major temperature changes within the workshop area.

Recommended actions

In order to remedy the symptom, the following actions are recommended (the actions are listed in order of probability):

	Action	Info/illustration
1.	Check all the cabling, especially the cables in the Emergency stop and Enable chains. Make sure all connectors are connected securely.	
2.	Check if any indication LEDs signal any malfunction that may give some clue to the problem.	The significance of all indication LEDs are specified in section *Indications on page 65*.
3.	Check the messages in the event log. Sometimes specific error combinations are intermittent.	The event log messages may be viewed either on the FlexPendant or using RobotStudio.
4.	Check the robot's behaviour, etc, each time that type of error occurs.	If possible, keep track of the malfunctions in a log or similar.
5.	Check whether any condition in the robot working environment also changes periodically, e.g, interference from any electric equipment only operating periodically.	
6.	Investigate whether the environmental conditions (such as ambient temperature, humidity, etc) has any bearing on the malfunction.	If possible, keep track of the malfunctions in a log or similar.

Continues on next page

4 Trouble shooting by Unit

4.1. Trouble shooting the FlexPendant

General

The FlexPendant communicates, through the Panel Board, with the Control Module main computer. The FlexPendant is physically connected to the Panel Board through a cable in which the +24 V supply and two Enabling Device chains run and emergency stop.

Procedure

The procedure below details what to do if the FlexPendant does not work correctly.

	Action	Info/illustration
1.	If the FlexPendant is completely "dead", proceed as detailed in section *Problem starting the FlexPendant on page 40*.	
2.	If the FlexPendant starts, but does not operate correctly, proceed as detailed in section *Problem connecting FlexPendant to the controller on page 41*.	
3.	If the FlexPendant starts, seems to operate, but displays erratic event messages, proceed as detailed in section *Erratic event messages on FlexPendant on page 42*.	
4.	Check the cable for connections and integrity.	
5.	Check the 24 V power supply.	
6.	Read the error event log message and follow any instructions of references.	

4.2. Trouble shooting communications

Overview

This section details how to trouble shoot data communication in the Control and Drive Modules.

Trouble shooting procedure

When trouble shooting communication faults, follow the outline detailed below:

	Action	Info/illustrations
1.	Faulty cables (e.g. send and receive signals are mixed up).	
2.	Transfer rates (baud rates).	
3.	Data widths that are incorrectly set.	

4.3. Trouble shooting fieldbuses and I/O units

Where to find information

Information about how to trouble shoot the fieldbuses and I/O units can be found in the manual for the respective fieldbus or I/O unit.

4.4 Trouble shooting power supply

4.4.1. Trouble shooting DSQC 604

Required test equipment

Equipment needed for trouble shooting:

- Ohmmeter
- Resistive load (e.g. Main Computer DSQC 639 on +24V_PC)
- Voltmeter

Preparations

	Action	Note
1.	Check the FlexPendant for errors and warnings.	

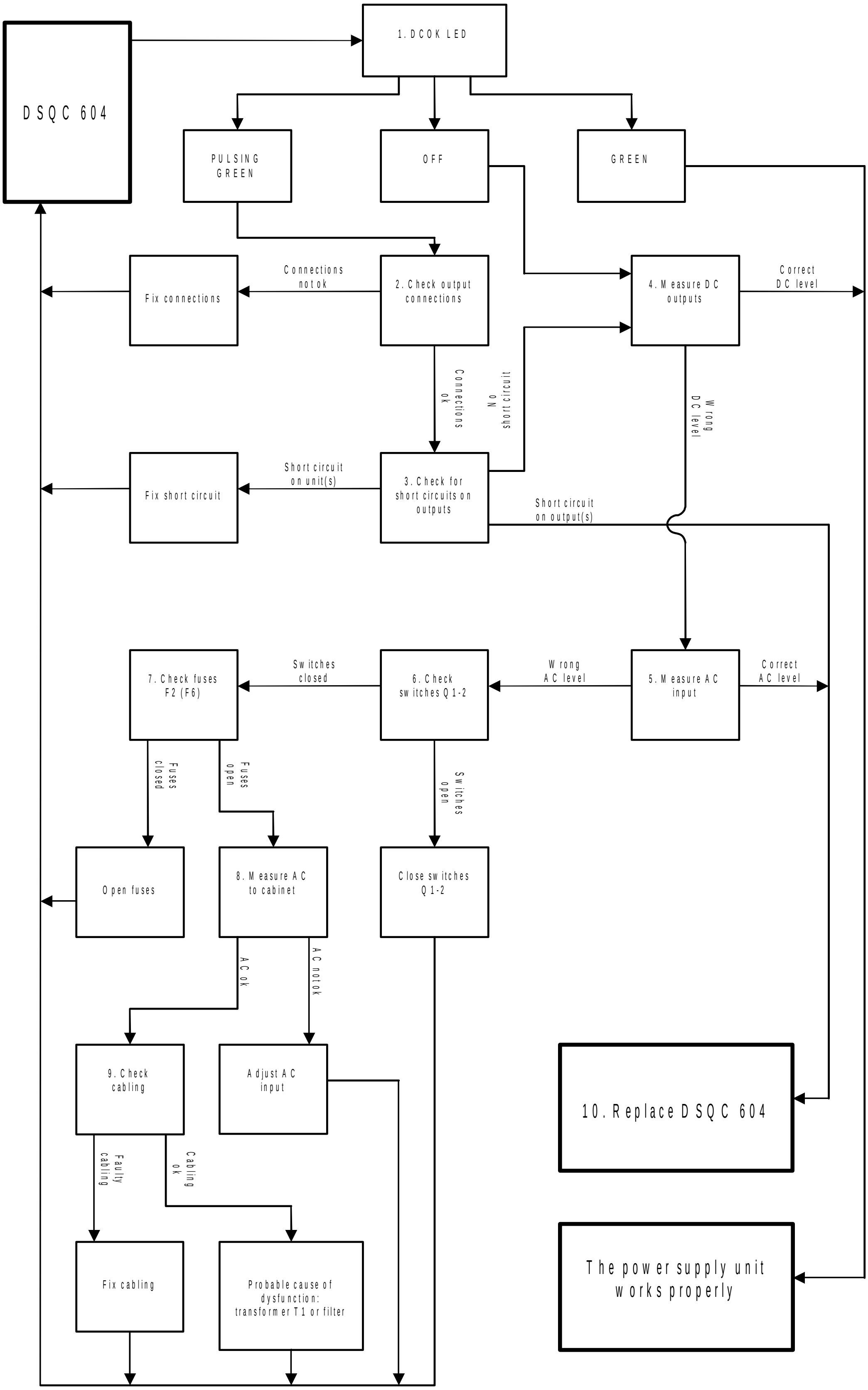
Trouble shooting procedure, DSQC 604

The trouble shooting table is supposed to be used as a detailed instruction together with the trouble shooting flowchart, see *Trouble shooting flowchart, DSQC 604 on page 58*.

	Test	Note	Action
1.	Check the indicator LED on DSQC 604.	The indicator LED is labelled DCOK.	If the LED is GREEN, the power supply should be working properly. If the LED is PULSING GREEN, the DC outputs are probably not connected to any units or there may be a short circuit on an output. Proceed with step 2. If the LED is OFF, either the power supply is faulty or it does not have sufficient input voltage. Proceed with step 4.
2.	Check connections between DC outputs and connected units.	Make sure that the power supply is connected to its proper units. A minimum load of 0.5-1A is required on at least one DC output for the 604 to work properly.	If the connections are OK, proceed with step 3. If the connections are faulty or the power supply is not connected to any units at all, repair connections/ connect units. Verify that the fault has been fixed and restart this guide if necessary.
3.	Check for short circuits on DC outputs.	Check both the DC outputs on DSQC 604 and the inputs on surrounding units. Measure the resistance between voltage pins and ground. The resistance should NOT be zero. The DC outputs are shown in the Circuit Diagram in *Product manual - IRC5*.	If no short circuit is found, proceed with step 4. If a short circuit is found on DSQC 604, proceed with step 10. If a short circuit is found on any surrounding unit, get that unit working. Verify that the fault has been fixed and restart this guide if necessary.

	Test	Note	Action
4.	Disconnect one DC output at a time and measure its voltage.	Make sure that at least one unit is connected at all times. A minimum load of 0.5- 1A is required on at least one output for the 604 to work properly. Measure the voltage using a voltmeter. The voltage should be: +24V < U < +27V. The DC outputs are shown in the Circuit Diagram in *Product manual - IRC5*.	If the correct voltage is detected on all outputs and the DCOK LED is green, the power supply is working properly. If the correct voltage is detected on all outputs and the DCOK LED is off, the power supply is regarded as faulty but does not have to be replaced instantly. If no or the wrong voltage is detected, proceed with step 5.
5.	Measure the input voltage to the 604.	Measure the voltage using a voltmeter. Voltage should be: 172 < U < 276V. The AC input is shown in the Circuit Diagram in *Product manual - IRC5*.	If the input voltage is correct, proceed with step 10. If no or the wrong input voltage is detected, proceed with step 6.
6.	Check switches Q1-2.	Make sure that they are closed. Their physical location is shown in the Circuit Diagram in *Product manual - IRC5*.	If the switches are closed, proceed with step 7. If the switches are open, close them. Verify that the fault has been fixed and restart this guide if necessary.
7.	Check main fuse F2 and optional fuse F6 if used.	Make sure that they are open. Their physical location is shown in the Circuit Diagram in *Product manual - IRC5*.	If the fuses are open, proceed with step 8. If the fuses are closed, open them. Verify that the fault has been fixed and restart this guide if necessary.
8.	Make sure that the input voltage to the cabinet is the correct one for that particular cabinet.		If the input voltage is correct, proceed with step 9. If the input voltage is incorrect, adjust it. Verify that the fault has been fixed and restart this guide if necessary.
9.	Check the cabling.	Make sure that the cabling is correctly connected and not faulty.	If the cabling is OK, the problem is likely to be the transformer T1 or the input filter. Try to get this part of the supply working. Verify that the fault has been fixed and restart this guide if necessary. If the cabling is found unconnected or faulty, connect/replace it. Verify that the fault has been fixed and restart this guide if necessary.

Trouble shooting flowchart, DSQC 604



en0800000354

4.4.2. Trouble shooting DSQC 661

Required test equipment

Equipment needed for trouble shooting:

- Ohmmeter
- Resistive load (e.g. Main Computer DSQC 639 on +24V_PC)
- Voltmeter

Preparations

	Action
1.	Check the FlexPendant for errors and warnings.
2.	Make sure that the control system power supply is in run-time mode. Do this by waiting 30 seconds after power-on.

Trouble shooting procedure, DSQC 661

The trouble shooting table is supposed to be used as a detailed instruction together with the trouble shooting flowchart, see *Trouble shooting flowchart, DSQC 661 on page 61*.

	Test	Note	Action
1.	Check the indicator LED on DSQC 661.	The indicator LED is labelled DCOK.	If the LED is GREEN, the 661 should be working properly. If the LED is PULSING GREEN, the DC output is probably not connected to any unit (load) or there may be a short circuit on the output. Proceed with step 2. If the LED is OFF, either the 661 is faulty or it does not have sufficient input voltage. Proceed with step 4.
2.	Check connection between DC output and connected unit.	Make sure that the power supply is connected to DSQC 662. A minimum load of 0.5-1A is required on the DC output for the 661 to work properly.	If the connection is OK, proceed with step 3. If the connection is faulty or the power supply is not connected to DSQC 662, repair connection/ connect it. Verify that the fault has been fixed and restart this guide if necessary.
3.	Check for short circuit on DC output.	Check both the DC output on DSQC 661 and the input on DSQC 662. Measure the resistance between voltage pins and ground. The resistance should NOT be zero. The DC output is shown in the Circuit Diagram in *Product manual - IRC5*.	If no short circuit is found, proceed with step 4. If a short circuit is found on DSQC 661, proceed with step 10. If a short circuit is found on DSQC 662, get that unit working. Verify that the fault has been fixed and restart this guide if necessary.

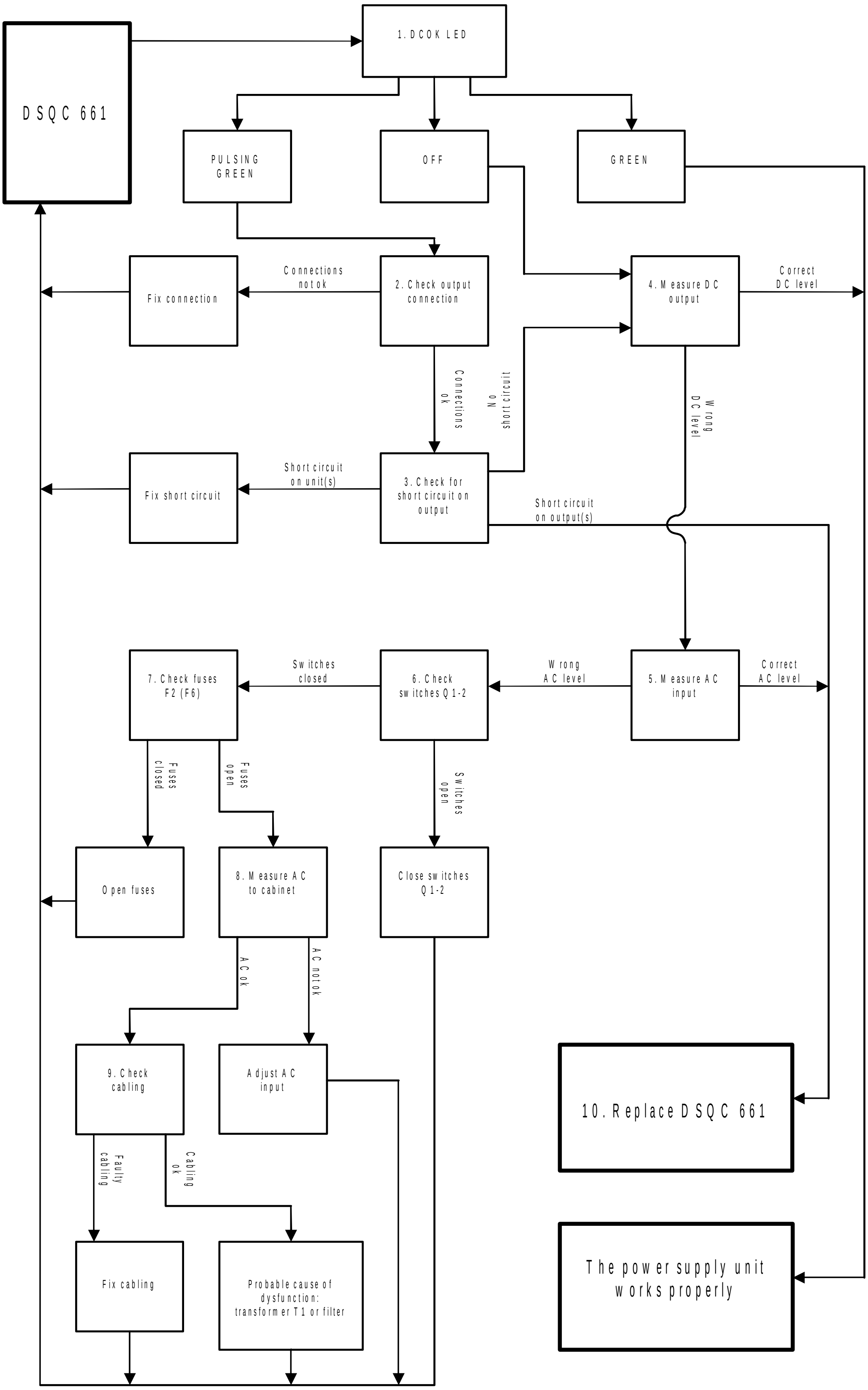
Continues on next page

Continued

	Test	Note	Action
4.	Measure the DC voltage while the output is connected to DSQC 662 or some other load.	DSQC 661 requires a minimum load of 0.5- 1A in order for it to deliver +24V. Measure the voltage using a voltmeter. The voltage should be: +24V < U < +27V. The DC output is shown in the Circuit Diagram in *Product manual - IRC5*.	If the correct voltage is detected and the DCOK LED is green, the power supply is working properly. If the correct voltage is detected and the DCOK LED is off, the power supply is regarded as faulty but does not have to be replaced instantly. If no or the wrong voltage is detected, proceed with step 5.
5.	Measure the input voltage to the 661.	Measure the voltage using a voltmeter. Voltage should be: 172 < U < 276V. The AC input is shown in the Circuit Diagram in *Product manual - IRC5*.	If the input voltage is correct, proceed with step 10. If no or the wrong input voltage is detected, proceed with step 6.
6.	Check switches Q1-2.	Make sure that they are closed. Their physical location is shown in the Circuit Diagram in *Product manual - IRC5*.	If the switches are closed, proceed with step 7. If the switches are open, close them. Verify that the fault has been fixed and restart this guide if necessary.
7.	Check main fuse F2 and optional fuse F6 if used.	Make sure that they are open. Their physical location is shown in the Circuit Diagram in *Product manual - IRC5*.	If the fuses are open, proceed with step 8. If the fuses are closed, open them. Verify that the fault has been fixed and restart this guide if necessary.
8.	Make sure that the input voltage to the cabinet is the correct one for that particular cabinet.		If the input voltage is correct, proceed with step 9. If the input voltage is incorrect, adjust it. Verify that the fault has been fixed and restart this guide if necessary.
9.	Check the cabling.	Make sure that the cabling is correctly connected and not faulty.	If the cabling is OK, the problem is likely to be the transformer T1 or the input filter. Try to get this part of the supply working. Verify that the fault has been fixed and restart this guide if necessary. If the cabling is found unconnected or faulty, connect/ replace it. Verify that the fault has been fixed and restart this guide if necessary.
10.	The 661 may be faulty, replace it and verify that the fault has been fixed.	How to replace the unit is detailed in *Product manual - IRC5*.	

Continues on next page

Trouble shooting flowchart, DSQC 661



en0800000355

4.4.3. Trouble shooting DSQC 662

Required test equipment

Equipment needed for trouble shooting:

- Ohmmeter
- Resistive load (e.g. Main Computer DSQC 639 on +24V_PC)
- Voltmeter

Preparations

	Action	Note
1.	Check the FlexPendant for errors and warnings.	
2.	Make sure that the power distribution board is in run-time mode. Do this by waiting 1 minute after power-on.	When the AC power has been cut off, the indicator LED (Status LED) on DSQC 662 will turn red and stay red until UltraCAP is empty. This may take a long time and is completely normal. It does not mean that there is something wrong with the 662.

Trouble shooting procedure, DSQC 662

The trouble shooting table is supposed to be used as a detailed instruction together with the trouble shooting flowchart, see *Trouble shooting flowchart, DSQC 662 on page 64*.

	Test	Note	Action
1.	Check the indicator LED on DSQC 662.	The indicator LED is labelled Status LED.	If the LED is GREEN, the 662 should be working properly. If the LED is PULSING GREEN, a USB communication error has occurred. Proceed with step 2. If the LED is RED, the input/output voltage is low, and/or the logic signal ACOK_N is high. Proceed with step 4. If the LED is PULSING RED, one or more DC outputs are under specified voltage level. Make sure cables are properly connected to its respective units. Proceed with step 4. If the LED is PULSING REDGREEN, a firmware upgrade error has occurred. This is not supposed to happen during runtime mode, proceed with step 6. If the LED is OFF, either the 662 is faulty or it does not have sufficient input voltage. Proceed with step 4.

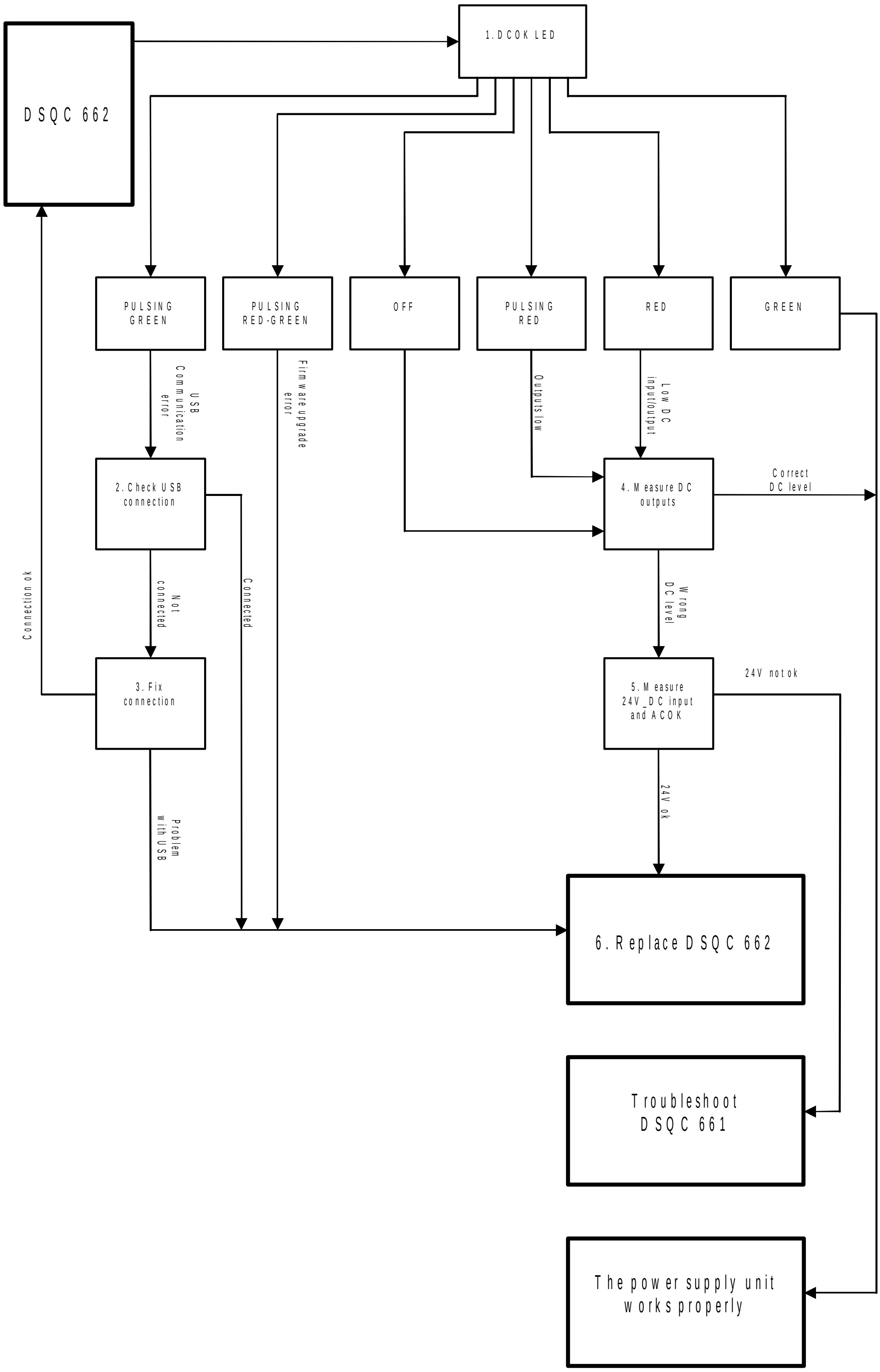
Continues on next page

Continued

	Test	Note	Action
2.	Check USB connection on both ends.		If the connection seems OK, proceed with step 6. If there is a problem with the connection, proceed with step 3.
3.	Try to fix the communication between the power supply and the computer by reconnecting the cable.	Make sure that the USB cable is properly connected on both ends.	If the communication comes back up, verify that the fault has been fixed and restart this guide if necessary. If unable to fix the communication, proceed with step 6.
4.	Disconnect one DC output at a time and measure its voltage.	Make sure that at least one unit is connected at all times. A minimum load of 0.5- 1A is required on at least one output for the 662 to work properly. Measure the voltage using a voltmeter. The voltage should be: +24V < U < +27V. The DC outputs are shown in the Circuit Diagram in *Product manual - IRC5*.	If the correct voltage is detected on all outputs and the Status LED is green, the power supply is working properly. If the correct voltage is detected on all outputs and the Status LED is NOT green, the power supply is regarded as faulty but does not have to be replaced instantly. If no or the wrong voltage is detected, proceed with step 5.
5.	Measure the input voltage to the 662 and the ACOK_N signal.	Measure the voltage using a voltmeter. Input voltage should be: 24 < U < 27V and ACOK_N should be 0V. Make sure that connectors X1 and X2 are connected properly on both ends. The DC input X1 and ACOK_N connector X2 are shown in the Circuit Diagram in *Product manual - IRC5*.	If the input voltage is correct, proceed with step 6. If no or the wrong input voltage is detected, troubleshoot DSQC 661.
6.	The 662 may be faulty, replace it and verify that the fault has been fixed.	How to replace the unit is detailed in *Product manual - IRC5*.	

Continues on next page

Trouble shooting flowchart, DSQC 662



en0800000356

5 Descriptions and background information

5.1 Indications

5.1.1. LEDs in the Control Module

General

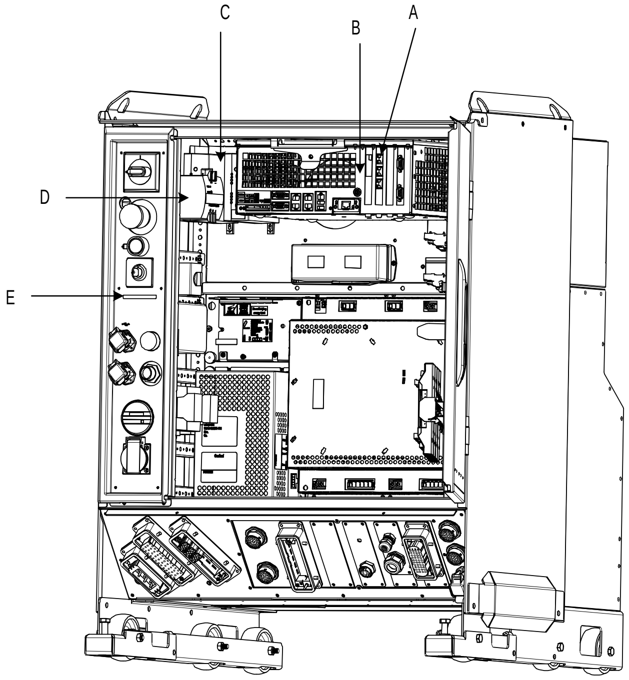
The Control Module features a number of indication LEDs, which provide important information for trouble shooting purposes. If no LEDs light up at all when switching the system on, trouble shoot as detailed in section *All LEDs are OFF at Controller on page 36*

All LEDs on the respective units, and their significance, are described in the following sections.

All units with LEDs are shown in the illustration below:

LEDs

Single cabinet controller

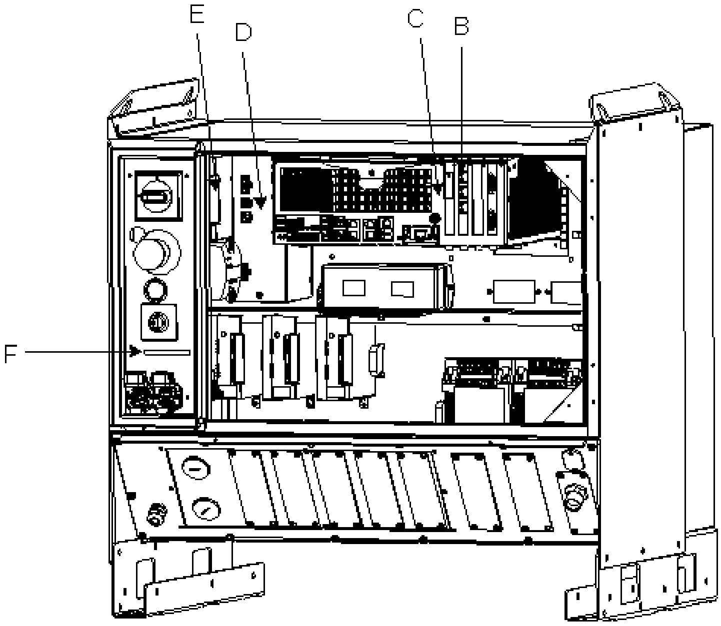


en1000000039

A	Ethernet board (any of the four board slots)
B	Computer unit (DSQC 639)
C	Customer I/O power supply (up to three units)
D	Control module power supply
E	LED board

Continues on next page

Control module for Dual cabinet controller



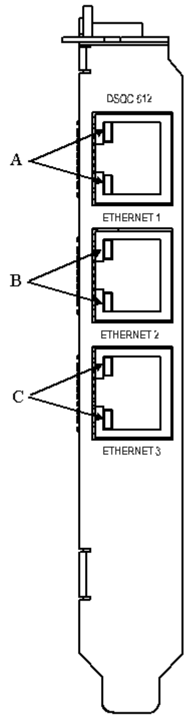
en1000000035

B	Ethernet board (any of the four board slots)
C	Computer unit (DSQC 639)
D	Customer I/O power supply (up to three units)
E	Control module power supply
F	LED board

Continues on next page

Ethernet board

The illustration below shows the LEDs on the Ethernet board:



en0400000919

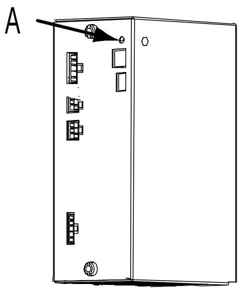
A	AXC2 connector LED
B	AXC3 connector LED
C	AXC4 connector LED

Description	Significance
AXC2 connector LED	Shows the status of Ethernet communication between Axis Computer 2 and the Ethernet board. GREEN OFF:10 Mbps data rate has been selected. GREEN ON:100 Mbps data rate has been selected. YELLOW flashing: The two units are communicating on the Ethernet channel. YELLOW steady: A LAN link is established. YELLOW OFF: A LAN link is *not* established.
AXC3 connector LED	Shows the status of Ethernet communication between Axis Computer 3 and the Ethernet board See the description above!
AXC4 connector LED	Shows the status of Ethernet communication between Axis Computer 4 and the Ethernet board See the description above!

Control module Power supply

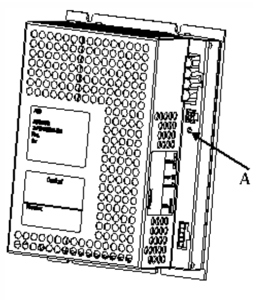
The illustration below shows the LEDs on the Control module power supply:

DSQC



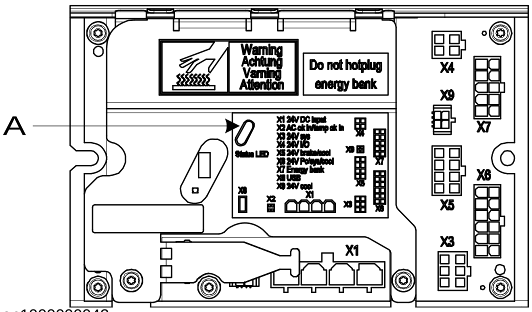
xx0400001073

DSQC 661



en1000000041

DSQC 662



en1000000042

A	DCOK indicator

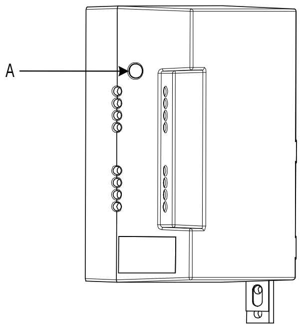
Description	Significance
DCOK indicator	GREEN: When all DC outputs are above the specified minimum levels. OFF: When one or more DC output/s below the specified minimum level.

Control module Power distribution board

Description	Significance
DCOK indicator	GREEN: When DC output is above the specified minimum level. OFF: When the DC output below the specified minimum level.

Customer Power Supply

The illustration below shows the LEDs on the Customer Power Supply Module:



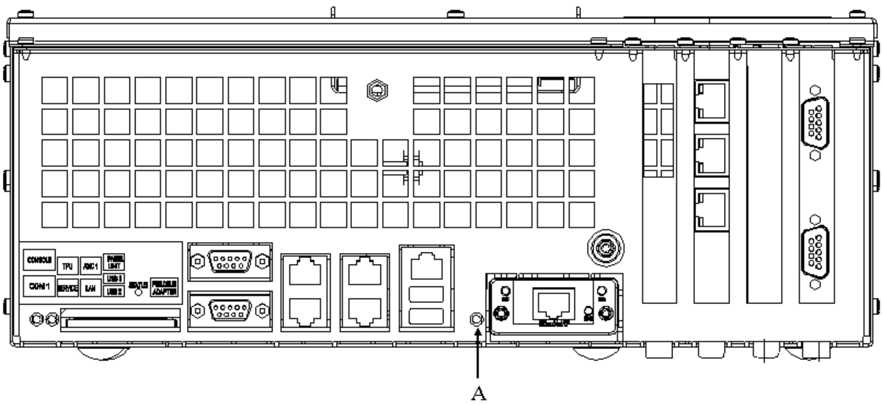
en1000000037

A	DCOK indicator

Description	Significance
DCOK indicator	GREEN: When all DC outputs are above the specified minimum levels. OFF: When one or more DC output/s below the specified minimum level.

Computer unit

The illustration below shows the LEDs on the Computer unit:



en1000000040

Description	Significance
Status LED	Shows the status of the communication on the computer unit

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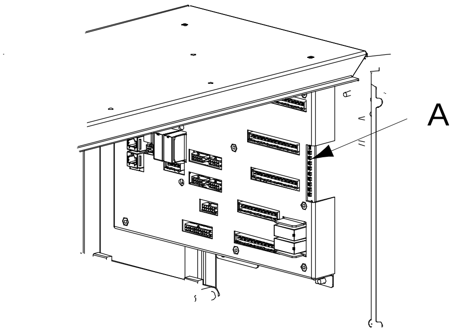
Panel board

The illustration below shows the LEDs on the Panel board:



dummy

A	Panel board LEDs

The panel board LEDs are described from top to bottom below:

Description	Significance
Status LED	GREEN flashing: serial communication error. GREEN steady: no errors found and system is running. RED flashing: system is in power up/selftest mode. RED steady: other error than serial communication error.
Indication LED, ES1	YELLOW when Emergency stop chain 1 closed
Indication LED, ES2	YELLOW when Emergency stop chain 2 closed
Indication LED, GS1	YELLOW when General stop switch chain 1 closed
Indication LED, GS2	YELLOW when General stop switch chain 2 closed
Indication LED, AS1	YELLOW when Auto stop switch chain 1 closed
Indication LED, AS2	YELLOW when Auto stop switch chain 2 closed
Indication LED, SS1	YELLOW when Superior stop switch chain 1 closed
Indication LED, SS2	YELLOW when Superior stop switch chain 2 closed
Indication LED, EN1	YELLOW when ENABLE1=1 and RS-communication is OK

LED board

The function of the LEDs on the LED board are identical to those on the Panel board as described above.

Should the LED board not be working, but the Panel board is, the problem is the communication between these boards or the LED board itself. Check the cabling between them.

Continues on next page

5.1.2. LEDs in the Drive Module for Drive System 04

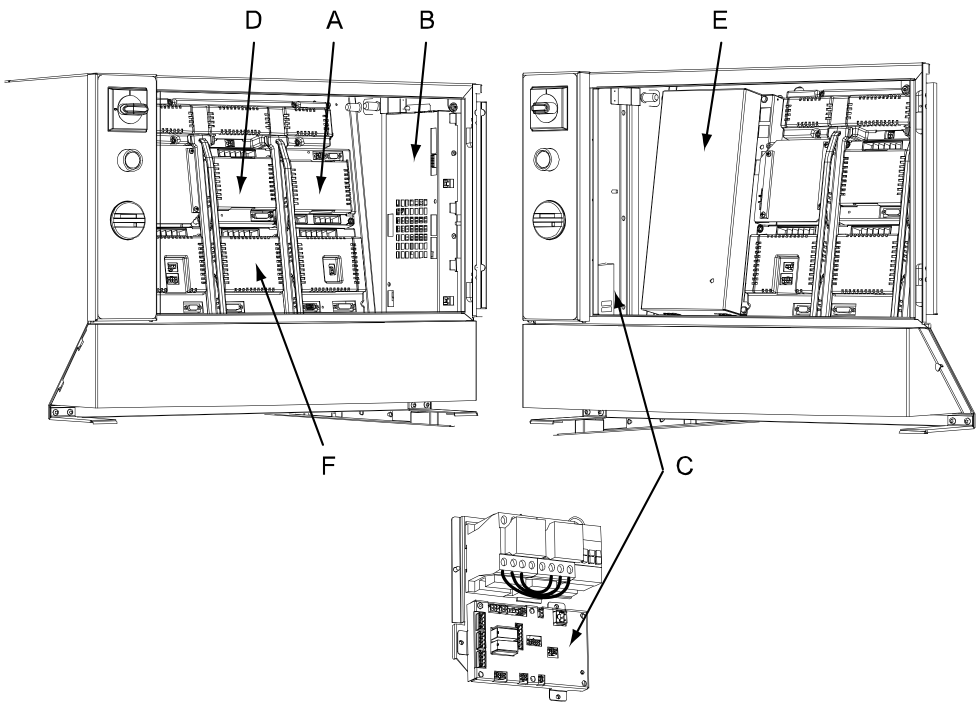
General

The Drive Module features a number of indication LEDs, which provide important information for trouble shooting purposes. If no LEDs light up at all when switching the system on, trouble shoot as detailed in section *All LEDs are OFF at Controller on page 36*.

All LEDs on the respective units, and their significance, are described in the following sections.

All units with LEDs are shown in the illustration below:

LEDs



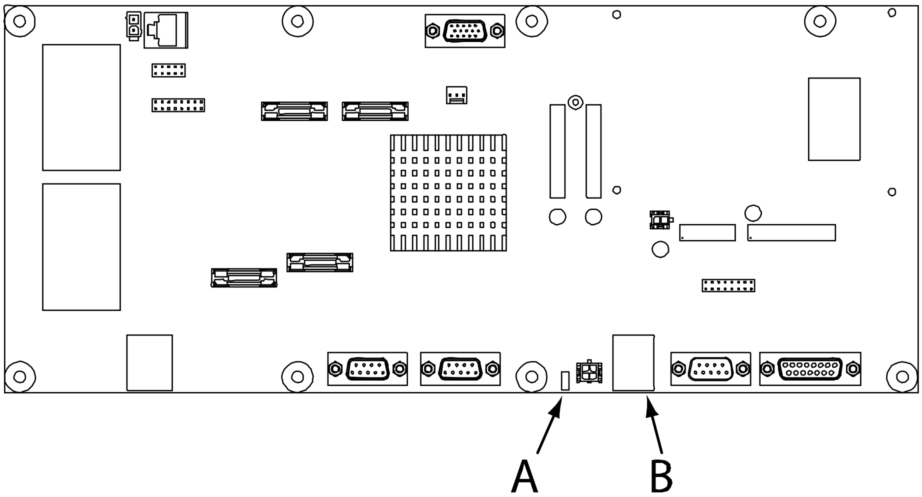
xx0400001084

A	Rectifier
B	Axis Computer
C	Contactor interface board
D	Single servo drive
E	Drive Module Power Supply
F	Main drive unit

Continues on next page

Axis computer

The illustration below shows the LEDs on the Axis computer:



xx0300000023

A	Status LED
B	Ethernet LEDs

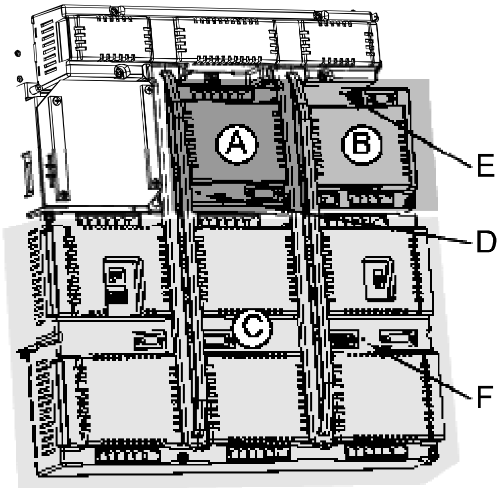
Description	Significance
Status LED	Normal sequence during startup: 1. RED steady: Default at power-up. 2. GREEN flashing: Establish connection to main computer, retrieve IP address and download the application file. 3. GREEN steady. Start-up sequence ready. Application is running. The following indicates errors: • RED steady (forever): The axis computer has failed to initialize basic hardware. • RED (long) -> GREEN flashing (short) -> RED (long) -> GREEN flashing (short) -> ... : Missing connection to main computer. Possible cable problem. • RED (short) -> GREEN flashing (long) -> RED (short) -> GREEN flashing (long) -> ... : Connection established with main computer. Possible RobotWare problem in main computer. • GREEN flashing (forever): Possible cable or RobotWare problem in main computer.
Ethernet LED	Shows the status of Ethernet communication between an additional axis computer (2, 3 or 4) and the Ethernet board. • GREEN OFF:10 Mbps data rate has been selected. This is an error state. • GREEN ON:100 Mbps data rate has been selected. • YELLOW flashing: The two units are communicating on the Ethernet channel. • YELLOW steady: A LAN link is established. • YELLOW OFF: A LAN link is *not* established.

Continues on next page

Main drive unit, single drive unit and rectifier unit

The illustration below shows the indication LEDs on the main drive unit, single drive unit and rectifier unit.

NOTE that there are two types of main drive units: a six unit drive and a three unit drive which are both used to power a six axes robot. Shown in the illustration is a six unit drive. The three unit drive is half the size of the six unit drive, but the indication LED is positioned in the same place.



xx0400001089

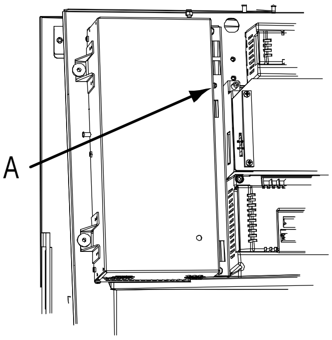
A	Single drive unit
B	Rectifier unit
C	Main drive unit
D	Indication LED, single servo drive
E	Indication LED, rectifier unit
F	Indication LED, main drive unit

Description	Significance
Indication LEDs D, E and F	GREEN flashing: Internal function is OK and there is a malfunction in the interface to the unit. The unit does not need to be replaced. GREEN steady: Program loaded successfully, unit function OK and all interfaces to the units is fully functional. RED steady: Permanent internal fault detected. The LED is to have this mode in case of failure at internal self test at start-up, or in case of detected internal failure state in running system. The unit probably needs to be replaced.

Continues on next page

Drive Module Power Supply

The illustration below shows the LEDs on the Drive Module power supply:



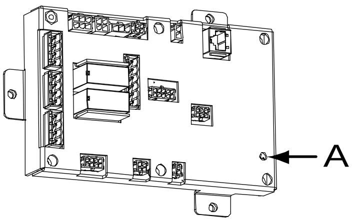
xx0400001090

A	DCOK indicator

Description	Significance
DCOK indicator	GREEN: When all DC outputs are above the specified minimum levels. OFF: When one or more DC output/s below the specified minimum level.

Contactor interface board

The illustration below shows the LEDs on the Contractor interface board:



xx0400001091

A	Status LED

Description	
Status LED	GREEN flashing: serial communication error. GREEN steady: no errors found and system is running. RED flashing: system is in power up/selftest mode. RED steady: other error than serial communication error.

5.1.3. LEDs in the Drive Module for Drive System 09

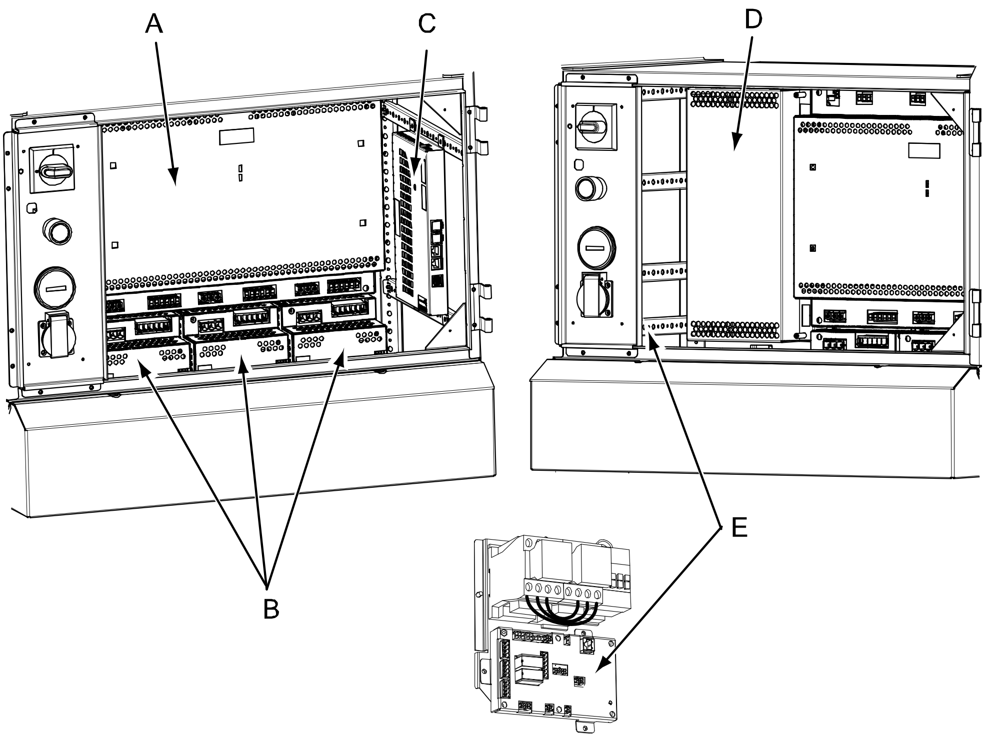
General

The Drive Module features a number of indication LEDs, which provide important information for trouble shooting purposes. If no LEDs light up at all when switching the system on, trouble shoot as detailed in section *All LEDs are OFF at Controller on page 36*.

All LEDs on the respective units, and their significance, are described in the following sections.

All units with LEDs are shown in the illustration below:

Units



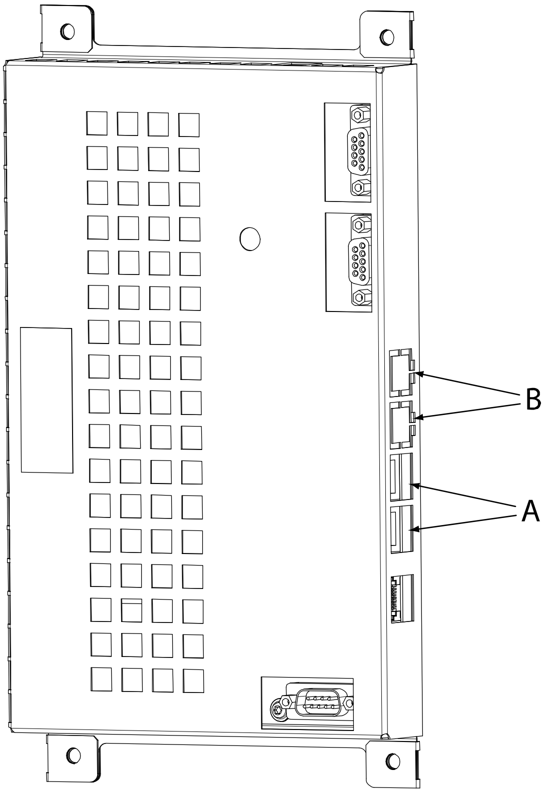
xx0800000484

A	Main Drive Unit
B	Additional Drive Units
C	Axis computer
D	Drive Module power supply
E	Contactor interface board

Continues on next page

Axis computer

The illustration below shows the LEDs on the Axis computer:



xx0800000485

A	Status LED
B	Ethernet LED

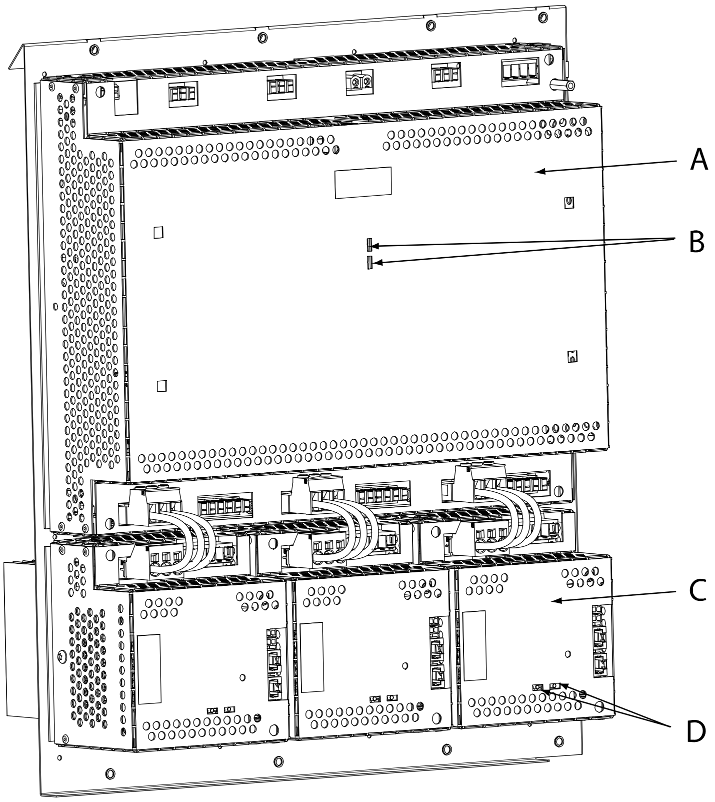
Description	Significance
Status LED	Normal sequence during startup: 1. RED steady: Default at power-up. 2. RED flashing: Establish connection to main computer and load program to axis computer. 3. GREEN flashing: Start-up of axis computer program and connect peripheral units. 4. GREEN steady. Start-up sequence ready. Application is running. The following indicates errors: • DARK: No power to axis computer or internal error (hardware/ firmware). • RED steady (forever): The axis computer has failed to initialize basic hardware. • RED flashing (forever): Missing connection to main computer, main computer start-up problem or RobotWare installation problem. • GREEN flashing (forever): Missing connections to peripheral units or RobotWare start-up problem.

Continues on next page

Description	Significance
Ethernet LED	Shows the status of Ethernet communication between an additional axis computer (2, 3 or 4) and the Ethernet board. • GREEN OFF:10 Mbps data rate has been selected. • GREEN ON:100 Mbps data rate has been selected. • YELLOW flashing: The two units are communicating on the Ethernet channel. • YELLOW steady: A LAN link is established. • YELLOW OFF: A LAN link is *not* established.

Main Drive Unit and Additional Drive Unit

The illustration below shows the indication LEDs on the Main Drive Unit and Additional Drive Units.



xx0800000486

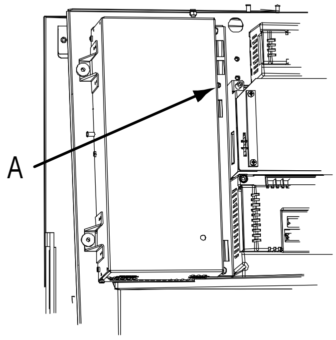
A	Main Drive Unit
B	Main Drive Unit Ethernet LEDs
C	Additional Drive Unit
D	Additional Drive Unit Ethernet LEDs

Continues on next page

Description	Significance
Ethernet LEDs (B and D)	Shows the status of Ethernet communication between an additional axis computer (2, 3 or 4) and the Ethernet board. • GREEN OFF:10 Mbps data rate has been selected. • GREEN ON:100 Mbps data rate has been selected. • YELLOW flashing: The two units are communicating on the Ethernet channel. • YELLOW steady: A LAN link is established. • YELLOW OFF: A LAN link is *not* established.

Drive Module Power Supply

The illustration below shows the LEDs on the Drive Module power supply:



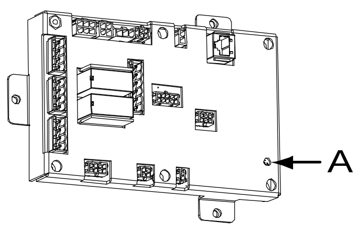
xx0400001090

A	DCOK indicator

Description	Significance
DCOK indicator	GREEN: When all DC outputs are above the specified minimum levels. OFF: When one or more DC output/s below the specified minimum level.

Contactor interface board

The illustration below shows the LEDs on the Contractor interface board:



xx0400001091

A	Status LED

Description	
Status LED	GREEN flashing: serial communication error. GREEN steady: no errors found and system is running. RED flashing: system is in power-up/self-test mode. RED steady: other error than serial communication error.